

Abstract

The ballistics helmet test produces massive amounts of sensor data which needs to be analyzed in order to assess biomechanical injury metrics including HIC36 and BrIC. However, the conventional method of analysis via the use of commercial software such as NI DIAdem is slow, expensive in terms of licensing fees, and lacks predictive ability.

The current project entails the development of a desktop application using the Python language for automating the biomechanical analysis process. The desktop application is developed using the PytS framework and it imports CSV and Excel files with sensor data (raw or filtered, computes the HIC36 and BrIC, finds the peak value of all twelve sensor channels, and shows the results in seven pairs of graphs. Another aspect of the current project included research into the Hybrid III head-neck assembly with the sensors installed on it.

In order to enhance the accuracy of the results generated by the Python software package, the generated results were compared against NI DIAdem with regard to the data acquired from nine ballistic tests. The corrections were made in the following manner, with a factor of 1.40 for HIC and 0.57 for BrIC being derived.

The software also incorporates a machine learning model for prediction that includes algorithms such as random forest, linear regression, and gradient boosting in order to predict the HIC and BrIC with respect to the speed of the projectile, projectile type, helmet type, and side of impact.

Other features included in the program are a graphical impact side selector, velocity versus drop height map, and injury severities.

In addition, a compact Prototype of a Projectile Velocity Measurement System (PVMS) was developed using a breadboard in order to obtain projectile velocities. These velocities were compared to the known reference values of the velocities obtained from the existing system and the percentage error was computed in order to test the accuracy of the prototype. The software also contains an impact side selection window, a drop height to velocity look-up table and injury severity categorization based on biomechanical standards.

The entire system can be summarized to provide an efficient platform for automated biomechanical analysis, velocity measurements, and injury predictions without expending much manual effort and still agreeing well with standard reference techniques.

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Chapter 1 - Introduction

1.1 Project Background

With the rising need for better personal protection in defense and law enforcement, ballistse helmets have come to be a vital part of protective gear. The key purpose of wearing such helmets is not just protection from penetration by projectiles, but also reduction in the transmission of impact forces to the wearer's head. Even if the projectile is stopped, the force that will be transmitted can be enough to cause TBI

Traumatic Brain Injury (TBI) refers to an injury caused by the action of an external mechanical force on the brain. TBI is divided into two types; that is, penetrating TBI, which involves a foreign body entering the brain, and non-penetrating or blunt TBI, involving rapid movement of the brain within the skull as a result of impact forces. In testing for helmet performance, the non-penetrating TBI is more of a problem since it could happen even if the helmet protects the head from projectile

penetration.

For the purpose of analyzing the efficiency of the ballistic helmet in resisting injuries caused by those mechanisms, a biomechanical test has to be conducted by means of an instrumented head form fitted with calibrated sensors. Data from the tests conducted on impact will be analyzed and injury criteria will be calculated; these criteria will be compared with safety levels.

1.2 Problem Definition

The current method of analyzing the collected data regarding the helmet impacts is based mainly on the use of commercially available software programs like NI DIAdem. This process involves manual loading of each individual CSV file, channel selection, filter setting, HIC and BrIC calculation, graph plotting, and writing of all these results down. All this has to be done for each separate test that makes this process quite lengthy and laborious. There is a high risk of errors when dealing with large number of tests performed on helmets using various ammunition types and different points of impact.

A third weakness of the present approach to the analysis is that there is no element of predictability within the system at all. The program only analyzes the results from tests carried out and cannot predict what the values of HIC or BrIC would be under various conditions of the speed of the projectile, the type of ammunition used, the type of helmet used, and the point of contact between the projectile and the helmet. This is especially useful considering the costliness of ballistic testing.

Morcover, the current process is not integrated for the velocity measurement of projectiles in the testing process. The measurement of velocity is always done separately through a commercially available chronograph and the value is then manually input increases the likelihood of data entry errors.

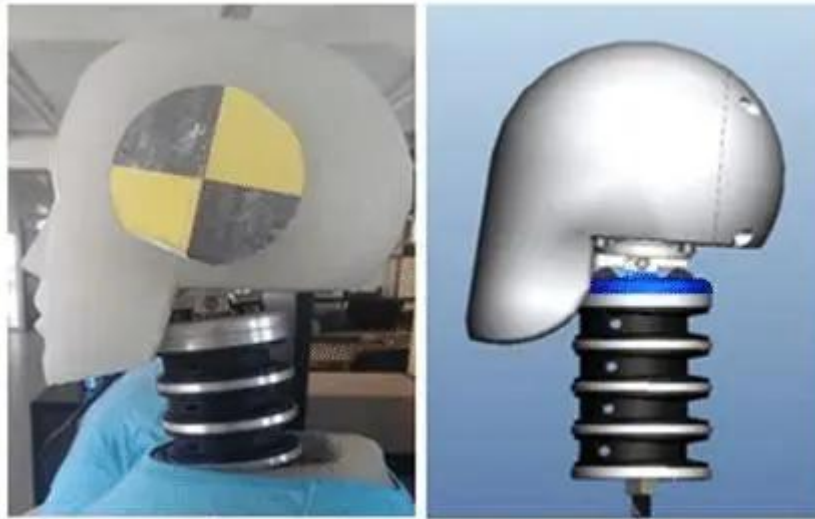


Figure 1: Head-Neck Assembly



Figure 2: Head-Neck Assembly Setup

1.3 Understanding the Hybrid III Head-Neck Assembly

Hybrid III 50th percentile male Anthropomorphic Test Device (ATD) is one of the most commonly employed dummy types for assessment of occupant restraints and protection devices. It was initially designed by General Motors and is presently supported by Humanetics in partnership with SAE International and NHTSA. Hybrid III ATD is listed in the United States Code of Federal Regulations (Part 572, Subpart E), and it is intended to reflect the dimensions of a typical adult male body.

The head module comprises a metal skull made from aluminum and coated with vinyl to mimic the shape of the human head. The neck module has alternating aluminum disks and rubber inserts that mimic the stiffness and damping characteristics of the human neck. This makes it possible for the headform to behave in response to the loads applied in a manner that is comparable to the way the human body behaves, thus allowing the recorded sensor information to be compared against known injury criteria for humans.

The sensors are placed on the head form at points that correspond to the centre of gravity of the human head. The linear accelerometers placed at this point sense acceleration in three mutually perpendicular directions. Angular rate sensors sense rotation around the three mutually perpendicular axes. The neck load cell is a six-axis sensor that measures force and moments transferred across the cervical spine of the human body.

1.4 Project Objectives

The main goals of the project include the following:

- Performing research on Hybrid III head-neck subsystem and getting familiar with the sensors employed by the system and their properties, including operating principle, channel number, measurement range, and unit of measurement.
- Creating a desktop application implemented with Python programming language, able to open the raw and filtered CSV/Excel data files, calculate the HIC36 and BrIC indicators, show peak values for all twelve sensor channels, and plot time-history graphs in pairs.
- Checking the differences between HIC and BrIC calculated by Python and NI DIAdem software, determining appropriate correction coefficients, and applying them to improve the accuracy of calculations.
- Implementing a machine learning prediction model which would be able to estimate HIC and BrIC indicators basing on parameters such as projectile velocity, ammunition type, helmet type, and impact side using a database of real experimental data and physics-generated data.
- Creating an intuitive interface that would contain a graphical head diagram for choosing the side of the impact, and a drop height table for easier input of velocities while making drop-test predictions.

Chapter 2 - Review of Literature

2.1 Head Injury Criterion (HIC)

The Head Injury Criterion (HIC) was formulated based on the work by Versace (1971) from the Biomechanics Research Laboratory at Wayne State University. This criterion has been devised based on earlier studies on the tolerance of head trauma, where it was found to have a correlation between the amount and duration of head acceleration and chances of skull fracture. The HIC is currently one of the most accepted biomechanical injury criteria in use today.

HIC combines both the **magnitude** and **duration** of head acceleration into a single injury metric, making it a more reliable indicator of head injury risk than peak acceleration alone. The **HIC36** criterion used in this project limits the maximum duration of the evaluation window to **36 milliseconds**, in accordance with the **SAE J211** and **FMVSS 208** standards.

HIC is defined as the maximum value of the following expression over all possible time intervals $[t_1, t_2]$:

$$HIC = (t_2 - t_1) \left(\left(\frac{1}{t_2 - t_1} \right) \int_{t_1}^{(t_2)} a(t) dt \right)^{2.5} \quad ()$$

where **a(t)** represents the resultant head acceleration measured in **g**, while **t₁** and **t₂** denote the start and end of the selected time interval in seconds.

The exponent **2.5** is based on experimental studies which showed that the severity of head injury depends not only on acceleration but also on the duration for which it acts. As a result, sustained high acceleration produces a much greater injury risk than a short-duration impact. Consequently, even small variations in the acceleration signal can significantly influence the calculated HIC value, highlighting the importance of appropriate signal filtering before analysis.

Algorithm: Sliding Window Maximisation

The HIC value is determined using a **brute-force sliding window algorithm** applied to the filtered resultant acceleration signal within a **±20 ms** interval around the peak acceleration. The computation is performed as follows:

- Select each sample within the impact window as the starting index (**i**).
- Extend the ending index (**j**) while ensuring that $(t[j] - t[i]) \leq 36 \text{ ms}$.
- Determine the average acceleration over the selected interval using **trapezoidal numerical integration**:

$$avg = \frac{\int a \, dt}{(t_2 - t_1)}$$

- Calculate the HIC value for that interval:

$$HIC \text{ candidate} = (t_2 - t_1) \times avg^{2.5}$$

- Repeat the process for all possible (**i, j**) combinations and record the **maximum HIC value** obtained.

At 50,000 Hz the impact window contains approximately 2,000 samples, making the $O(n^2)$ complexity practical for processing an individual test file.

Table 1: HIC Risk Classification

HIC Range	Risk Level	Standard Reference
0 – 250	Safe	EN 397 drop test safe limit
250 – 1000	Moderate	NIJ 0106.01 ballistic pass/fail threshold
1000 – 2400	Severe	FMVSS 208 severe injury region
> 2400	Critical	FMVSS 208 fatal injury threshold

2.2 Brain Injury Criterion (BrIC)

Takhounts et al. (2013) proposed Brain Injury Criterion (BrIC) to address the drawback of HIC, since HIC only accounts for translational acceleration but ignores brain injury due to rotational acceleration. An important mechanism of injury because of rotational acceleration is Diffuse Axonal Injury (DAI). DAI happens when the brain moves rotationally within the skull, stretching and shearing the white matter axons. This injury may lead to unconsciousness, cognitive dysfunction, and, in severe cases, brain injury or even death.

BrIC is defined as:

$$BrIC = \sqrt{\left(\frac{\omega_x}{W_{XC}}\right)^2 + \left(\frac{\omega_y}{W_{YC}}\right)^2 + \left(\frac{\omega_z}{W_{ZC}}\right)^2}$$

where:

- ω_x , ω_y , ω_z are the **peak angular velocities** about the **X (Roll)**, **Y (Pitch)**, and **Z (Yaw)** axes, respectively, measured in **rad/s**.
- **WXC**, **WYC**, and **WZC** represent the critical angular velocity constants corresponding to a **50% probability of AIS 4+ brain injury**. These constants were established by **Takhounts et al.** using reconstructed accident data and finite element head model simulations.

The critical angular velocity constants used are:

- **WXC = 66.25 rad/s** (*Roll – highest tolerance*)
- **WYC = 56.45 rad/s** (*Pitch*)
- **WZC = 42.87 rad/s** (*Yaw – lowest tolerance and the most critical axis*)

Algorithm: Peak Detection on Resultant Time Series

Unlike HIC, BrIC does not require a sliding window. The algorithm is:

- Convert from deg/s to rad/s: $\omega_{rad} = \omega_{deg} \times \frac{\pi}{180}$
- Compute BrIC at every sample: $BrIC(t) = \sqrt{\left(\frac{\omega_x}{W_{XC}}\right)^2 + \left(\frac{\omega_y}{W_{YC}}\right)^2 + \left(\frac{\omega_z}{W_{ZC}}\right)^2}$

- The **peak BrIC** is obtained by selecting the **maximum BrIC value** over the entire duration of the test.

Table 2: BrIC Risk Classification

BrIC Range	Risk Level	Description
0 – 0.20	Safe	No brain injury expected
0.20 – 0.45	Low	Minor rotational stress
0.45 – 0.65	Moderate	Moderate DAI risk
0.65 – 0.80	Severe	Severe DAI probable
> 0.80	Critical	Catastrophic brain injury

2.3 Helmet Testing Standards

Three standards are relevant to this project:

- **NIJ 0106.01 (National Institute of Justice, USA):** This is the primary standard used for the certification of ballistic helmets. It specifies that the **HIC value must not exceed 1000** during the acceptance testing procedure. Based on the level of protection offered against different ammunition types, helmets are classified into protection levels ranging from **Level I to Level IIIA**.
- **EN 397 (European Standard for Industrial Safety Helmets):** This standard applies to the performance evaluation of industrial safety helmets under drop-test conditions. Owing to the comparatively lower impact energies involved, the acceptable **HIC limit is 250**, which is considerably lower than that specified for ballistic helmet testing.
- **IS 4151 (Bureau of Indian Standards):** This Indian standard specifies the requirements for protective helmets and serves as the national reference for the procurement and evaluation of helmets used by defence and paramilitary organisations.

2.4 Existing Tools and Research Gap

NI DIAdem is one of the most widely used commercial software tools for biomechanical analysis of helmet impact data. It provides built-in functions for calculating **HIC** and **BrIC**, configurable signal filtering options, graphical visualization, and automated report generation. Despite these capabilities, the software requires a commercial licence and cannot be readily deployed on systems without a licensed installation. Furthermore, it does not offer any facility for predicting injury metrics before conducting physical impact tests.

Another commonly used platform is **MATLAB** with the **Signal Processing Toolbox**, which is extensively employed for signal processing and biomechanical data analysis in research applications. Although MATLAB offers powerful computational and visualization features, its usage is also restricted by licensing requirements.

A review of the available literature indicates that there is no publicly available, licence-free **Python-based** application that integrates automated biomechanical analysis with **machine learning-based injury prediction** into a single platform. This identified research gap served as the motivation for developing the system presented in this project, which combines automated signal processing, injury metric computation, and predictive analysis within a unified application.

Chapter 3 - Sensors and Experimental Setup

3.1 Overview of the Head-Neck Assembly

Impact testing of all helmets was performed using a Hybrid III 50th percentile male Anthropometric Test Device (ATD) head-neck assembly. The headform consists of an aluminum skull covered with vinyl skin, closely replicating the geometry and mechanical properties of a human head. The neck is constructed from aluminum and rubber discs designed to mimic human neck flexion and extension. Figure 3 shows the complete Hybrid III head-neck assembly, while Figure 4 illustrates the six-axis neck load cell mounting.

The assembly is equipped with three types of sensors: linear accelerometers, angular rate sensors, and a six-axis neck load cell. Linear accelerometers measure acceleration along three perpendicular axes, angular rate sensors capture rotational motion, and the load cell records forces and moments acting on the neck.

Sensor outputs are recorded using a data acquisition system at a sampling rate of **50,000 samples per second per channel**. Each impact test generates a CSV file containing a 22-row metadata header, a time column, and 12 sensor data channels.



Figure 3: Load Cell - Six-Axial, Mounted on Headform



Figure 4: Load Cell Assembly – Middle View

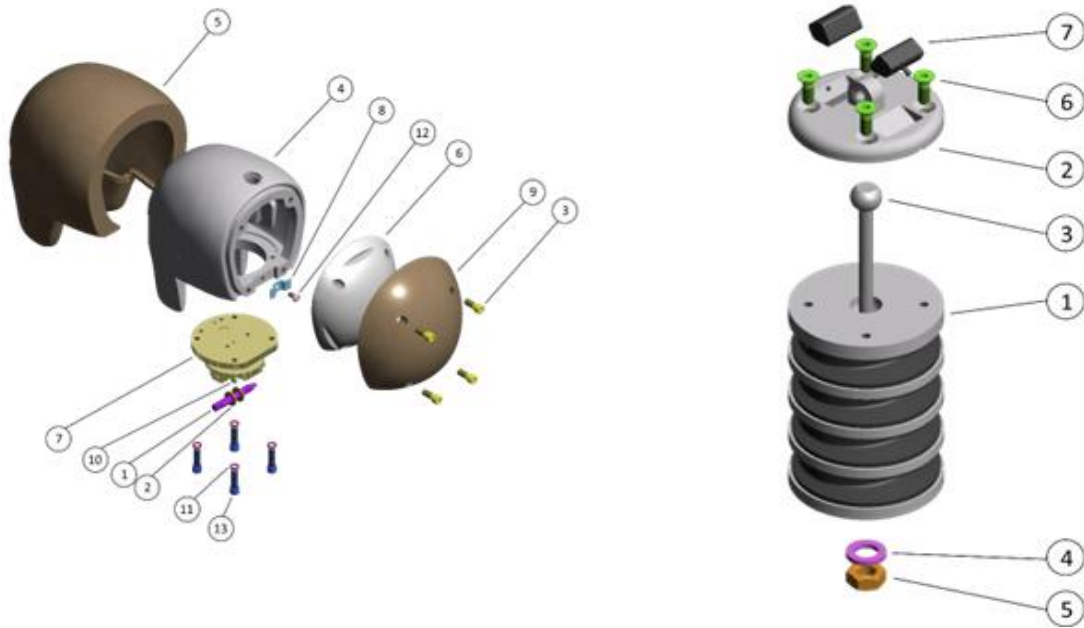


Figure 4: Head and Neck Assembly - Exploded Views

3.2 Linear Acceleration Sensors (Endevco T-Series)

Three piezoresistive accelerometers, which provide measurement of translational acceleration along the X, Y, and Z axes, are installed in the Hybrid III headform. These instruments are attached to the centre of mass of the headform to register the translational acceleration produced during the collision. The working principle of piezoresistive accelerometers is based on the piezoresistive effect in which the seismic mass produces strain in silicon flexure beams with embedded piezoresistors.

The resistance of the piezoresistive sensors changes as a result of the accelerating force acting on the sensor in proportion to the loading force. This change in resistance is measured by a Wheatstone bridge circuit, and as a result, an output voltage is produced which is directly proportional to the acceleration.

Table 3: Linear Accelerometer Channel Assignments

Channel	Sensor ID	Axis	Unit	Range
Chan 0	T31542	X (fore-aft)	g	± 2000 g
Chan 1	T32923	Y (lateral)	g	± 2000 g
Chan 2	T31139	Z (vertical)	g	± 2000 g

Personal Observation

During my study on the sensor layout, I noticed that for successful detection of linear acceleration, it is necessary to arrange each uniaxial accelerometer properly inside the headform.

Table 4: Endevco 7268C Accelerometer Specifications

Parameter	7268C-2000 (Value)	7268C-500 (Value)	Source
Measurement Range	±2000 g	±500 g	Endevco Datasheet 083019
Typical Sensitivity (10V excitation)	0.20 mV/g (typ)	0.80 mV/g (typ)	Endevco Datasheet 083019
Minimum Sensitivity	0.15 mV/g (min)	0.50 mV/g (min)	Endevco Datasheet 083019
Frequency Response (Z-axis)	0 – 3,000 Hz (±5%)	0 – 3,000 Hz (±5%)	Endevco Datasheet 083019
Frequency Response (X,Y axes)	0 – 1,500 Hz (±5%)	0 – 1,500 Hz (±5%)	Endevco Datasheet 083019
Mounted Resonance Freq. (Z)	26,000 Hz	17,000 Hz	Endevco Datasheet 083019
Damping Ratio	0.05 of critical	0.05 of critical	Endevco Datasheet 083019
Non-linearity + Hysteresis	±1% FSO	±1% FSO	Endevco Datasheet 083019
Shock Survival (half-sine)	10,000 g (200 μs)	5,000 g (300 μs)	Endevco Datasheet 083019
Transverse Sensitivity	<3% max	<3% max	Endevco Datasheet 083019
Zero Measurand Output	±100 mV max	±100 mV max	Endevco Datasheet 083019
Excitation Voltage	5 or 10 Vdc	5 or 10 Vdc	Endevco Datasheet 083019
Weight (sensor only)	8 g	8 g	Endevco Datasheet 083019
Dimensions	12.7 × 10.7 × 13.7 mm	12.7 × 10.7 × 13.7 mm	Endevco PR Release, 2010
Operating Temperature	–18°C to +66°C	–18°C to +66°C	Endevco Datasheet 083019
Calibration Traceability	NIST traceable	NIST traceable	Endevco Datasheet 083019

Datasheet: Endevco 7268C specifications obtained from the official Endevco datasheet.

3.3 Angular Rate Sensors (ARS)

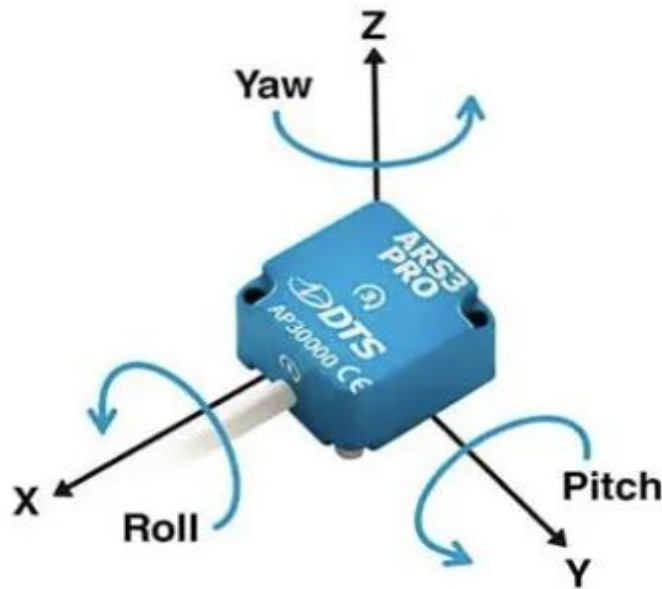
Three uniaxial ARSs are used for measuring the angular rotation about the X (roll), Y (pitch), and Z (yaw) axes. The ARSs work based on the Coriolis principle, which states that the vibration component of the sensor experiences a force acting perpendicular to the vibration as well as the rotating axis. This Coriolis force is directly proportional to the angular velocity of the input, thus resulting in an output voltage that is proportional to the rotational speed in deg/s.

The channel assignments, sensor IDs, measurement axes, and output units of the angular rate sensors used during testing are listed in Table 5.

Table 5: Angular Rate Sensor Channel Assignments

Channel	Sensor ID	Axis	Unit
Chan 3	ARS21556	X - Roll	deg/s
Chan 4	ARS15978	Y - Pitch	deg/s
Chan 5	ARS21564	Z - Yaw	deg/s

The orientation of the **roll (X)**, **pitch (Y)**, and **yaw (Z)** axes with respect to the headform is illustrated in **Figure 6**. Understanding this axis convention is essential for correctly interpreting the angular velocity measurements used in BrIC computation.

*Figure 5: Angular Velocity - Sensor Axis Convention*

Since the **BrIC** calculation requires angular velocity in **radians per second (rad/s)**, the measured values must first be converted from **degrees per second (deg/s)** using:

$$\omega_{rad} = \omega_{deg} \times \frac{\pi}{180}$$

During the initial implementation, this conversion was unintentionally omitted, which resulted in BrIC values approximately **57 times higher** than the correct values. The issue was later identified and corrected with the guidance of the project mentor.

The detailed specifications of the **DTS ARS3 PRO** sensors used in this study are summarized in **Table 6**.

Table 6: DTS ARS3 PRO Angular Rate Sensor Specifications

Parameter	ARS3 PRO-1500 Value	ARS3 PRO-8K Value	Source
Range	±1,500 deg/sec	±8,000 deg/sec	DTS Datasheet DSH-019, Rev. 12.2017
Bandwidth (standard)	0–2,000 Hz	0–2,000 Hz	DTS Datasheet DSH-019, Rev. 12.2017
Noise (rated BW)	<0.15% FSO	<0.30% FSO	DTS Datasheet DSH-019, Rev. 12.2017
Non-Linearity	<0.5% full scale	<0.5% full scale	DTS Datasheet DSH-019, Rev. 12.2017
Cross-Axis Sensitivity	<1.0%	<1.0%	DTS Datasheet DSH-019, Rev. 12.2017
Shock Survival	10,000 g, 0.5 ms	10,000 g, 0.5 ms	DTS Datasheet DSH-019, Rev. 12.2017
IP Rating	IP67	IP67	DTS Datasheet DSH-019, Rev. 12.2017
Excitation Voltage	5.0–14.0 Vdc	5.0–14.0 Vdc	DTS Datasheet DSH-019, Rev. 12.2017
Full Scale Output	±2 V nominal	±2 V nominal	DTS Datasheet DSH-019, Rev. 12.2017
Weight	10 g (0.35 oz)	10 g (0.35 oz)	DTS Datasheet DSH-019, Rev. 12.2017
Dimensions	19 × 19 × 12.5 mm	19 × 19 × 12.5 mm	DTS Datasheet DSH-019, Rev. 12.2017
Operating Temp.	–40 to +85°C	–40 to +85°C	DTS Datasheet DSH-019, Rev. 12.2017
Calibration	NIST traceable; ISO 17025 A2LA accredited	NIST traceable; ISO 17025 A2LA accredited	DTS Datasheet DSH-019, Rev. 12.2017

Datasheet: DTS ARS3 PRO specifications from the official datasheet DSH-019, Rev. 12.2017.

3.4 Six-Axis Neck Load Cell (Denton EW7402N)

The **Denton EW7402N six-axis neck load cell** is installed between the **base of the skull** and the **upper cervical spine** of the Hybrid III head-neck assembly. It measures **three orthogonal force** and **three orthogonal moment** components during impact.

The load cell uses **strain gauges** arranged in **Wheatstone bridge circuits**. Applied forces produce strain, causing a change in electrical resistance that is converted into voltage signals proportional to the measured forces and moments.



Figure 6: Load Cell - Six-Axial

Table 7: Load Cell – Six-Axial

Channel	Sensor ID	Measurement	Unit
Chan 6	EW7402N-MX	Moment X	Nm
Chan 7	EW7402N-MY	Moment Y	Nm
Chan 8	EW7402N-MZ	Moment Z	Nm
Chan 9	EW7402N-FX	Force X	N
Chan 10	EW7402N-FY	Force Y	N
Chan 11	EW7402N-FZ	Force Z	N

	Unit	Value					
		F _x	F _y	F _z	M _x	M _y	M _z
Measuring range	kN Nm	8.9	8.9	13.5			
Sensitivity ¹⁾	μV/V/kN μV/V/Nm	180	170	90	5.3	5.3	7.8
Output signal ^{1), 2)}	mV/V	1.6	1.5	1.2	1.5	1.5	2.2
Bridge resistance	Ω	350	350	700	350	350	700
Zero signal ¹⁾	mV/V	≤ 0.05					
Amplitude non-linearity ³⁾	%	≤ 1.0					
Hysteresis ³⁾	%	≤ 1.0					
Channel crosstalk ³⁾	%	≤ 5.0					
Supply voltage	V	2–15					
Ultimate load	%	150					
Insulation resistance	MΩ	> 100					
Temperature range	°C	-30...+70					
Weight (approximate)	g	630					

Figure 7: Six-Axis Neck Load Cell - Denton EW7402N

3.5 Test Types and Conditions

The experimental evaluation was carried out under two different impact conditions: ballistic tests and drop tests. Each test was designed to evaluate helmet performance under specific loading conditions.

Ballistic Tests

During ballistic testing, live ammunition was fired from a **gun barrel stand** towards the helmet mounted on the Hybrid III headform. The experimental setup ensured a fixed **standoff distance** and **impact angle**, providing consistent test conditions across all trials. Projectile velocity was measured using a **commercial chronograph** positioned between the muzzle and the target.

The ballistic tests were conducted using the following ammunition:

- **9 mm (Full Metal Jacket)** – lower mass with a moderate velocity of approximately **380–450 m/s**.
- **7.62 × 39 mm (AK-47 pattern)** – intermediate mass with a higher velocity of approximately **600–780 m/s**.

Drop Tests

Drop tests were performed to assess helmet performance under **low-velocity blunt impacts**, representing conditions such as accidental falls or industrial impacts. In these tests, the helmeted headform was released from predetermined heights onto a flat anvil surface. The corresponding impact velocity was calculated using the conservation of energy relationship:

$$v = \sqrt{2gh}$$

The impact velocities corresponding to the selected drop heights are listed in Table 8.

Table 8: Drop Height to Impact Velocity Conversion

Drop Height (m)	Impact Velocity (m/s)
0.5	3.13
1.0	4.43
1.5	5.42
2.0	6.26

Following the drop tests, these reference velocities were incorporated into the prediction module through a drop-height selection feature, eliminating the need for manual velocity calculations.

Helmet Types Tested

Table 9: Helmet Types Used in Testing

Helmet Type	Description	Test Type
Small (9mm)	9mm-rated ballistic helmet - small size	Ballistic, Drop
Medium (9mm)	9mm-rated ballistic helmet - medium size	Ballistic
Large (9mm)	9mm-rated ballistic helmet - large size	Ballistic, Drop
MSC	Multi-Shot Capability - higher protection	Ballistic, Drop
Without Helmet	Bare headform - baseline reference	Both

3.6 Data Acquisition

The sensor signals from the impact tests were obtained through a multichannel data acquisition device that was able to record the data from all sensors simultaneously at a sampling rate of 50,000 Hz per channel. The high sampling rate ensured that the short-lived phenomena arising from ballistics and drops could be accurately captured.

Each test yielded about five seconds of recordings, which created data files of almost 250,000 samples each. The recorded data were saved in CSV/excel format, starting from 22 lines of meta data which contained the information on calibration, channels used, and units of measurement.

After the meta data, there were a header line consisting of the Time line and the names of the twelve channels, followed by the sensor data. The overall organization of the data file is illustrated in Figure 9.

Headers												
Test Date	1/1/2026	1/1/2026	1/1/2026	1/1/2026	1/1/2026	1/1/2026	1/1/2026	1/1/2026	1/1/2026	1/1/2026	1/1/2026	1/1/2026
Test Time	14:33:25	14:33:25	14:33:25	14:33:25	14:33:25	14:33:25	14:33:25	14:33:25	14:33:25	14:33:25	14:33:25	14:33:25
Test ID	TBR-023_£5m_R	TBR-023_£5m_R	TBR-023_£5m_R	TBR-023_£5m_R	TBR-023_£5m_R	TBR-023_£5m_R	TBR-023_£5m_R	TBR-023_£5m_R	TBR-023_£5m_R	TBR-023_£5m_R	TBR-023_£5m_R	TBR-023_£5m_R
Test Description												
Sample Ra	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000
Hardware	10000	10000	10000	10000	10000	10000	10000	10000	10000	10000	10000	10000
Data Chan	1	2	3	4	5	6	7	8	9	10	11	12
ISO Chann	????????	????????	????????	????????	????????	????????	????????	????????	????????	????????	????????	????????
Channel D	Uni-axial a	Uni-axial A	Uni-axial A	Uni-axial A	Uni-axial A	Uni-axial A	Upper nec	Upper nec	Upper nec	Upper nec	Upper nec	Upper neck load cell
Channel Lc	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
Sensor S/N	T31542	T32923	T31139	ARS21556	ARS15978	ARS21564	EW7402N	EW7402N	EW7402N	EW7402N	EW7402N	EW7402N-FZ
Software f	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000
Software f	1650	1650	1650	1650	1650	1650	1650	1650	1650	1650	1650	1650
Engineerin G	G	G	G	Degree/Se	Degree/Se	Degree/Se	Nm	Nm	Nm	N	N	N
User Comment												
Number of	0	0	0	0	0	0	0	0	0	0	0	0
Number of	253041	253041	253041	253041	253041	253041	253041	253041	253041	253041	253041	253041
Data Zero	-318	-32768	-32768	-36	-9	-20	-64	-36	-97	-6	-91	-69
Scale Fact	-0.09707	-0.09681	-0.09653	-0.82402	-0.82297	-0.81165	0.016616	-0.01667	-0.01169	0.507073	-0.5217	0.492465
Scale Fact	-0.019	-0.01903	-0.0019	-0.07569	-0.07589	-0.07561	-0.00095	-0.00096	-0.00095	-0.00095	-0.00096	-0.00048
Data Starts Here												
Time	Chan 0:T3	Chan 1:T3	Chan 2:T3	Chan 3:AR	Chan 4:AR	Chan 5:AR	Chan 6:EW	Chan 7:EW	Chan 8:EW	Chan 9:EW	Chan 10:E	Chan 11:EW7402N-FZ

Figure 8: Data Acquisition Hardware

Chapter 4 - Software Architecture

4.1 Technology Selection

The following three main reasons prompted choosing Python as the main programming language for this application development.

First, this project required an off-line application that should work independently of proprietary software packages. Python language along with scientific computing software including such libraries as NumPy, SciPy, Pandas, and Matplotlib possesses all necessary features for signal processing and data analysis, remaining at the same time completely open-source.

Secondly, Python was already known due to academic education courses, and hence the process of learning the programming language became faster and easier, leaving more time for solving the problems of signal processing in biomechanics.

Thirdly, PyQt5 package was chosen for creating GUI of the application since it allows developing professional desktop applications, enabling the creation of convenient and efficient to use interface that can be used by people having no programming skills.

Used in this application Python libraries are listed in Table 10.

Table 10: Python Libraries and Their Role in the Application

Purpose	Library	Key Use in This Project
Desktop UI	PyQt5	Main window, tabs, buttons, card widgets, graph canvases
Signal filtering	SciPy	Butterworth filter (butter, filtfilt), numerical gradient
Numerical computing	NumPy	Array operations, trapezoidal integration, deg2rad
Data loading	Pandas	CSV/Excel parsing, column type coercion
Graphing	Matplotlib	12 embedded time-history plots, HIC/BrIC zoom panels
Machine learning	Scikit-learn	Random Forest, Regression, Gradient Boosting
Model persistence	Joblib	Save and load trained model bundle as .pkl file
PDF export	ReportLab	Multi-page PDF report with graphs and metrics

Development Approach

During the development process, the algorithms such as signal filtering, HIC calculation, BrIC calculation, and peak value calculation were first developed and tested through Jupyter notebook. This was done to check the intermediate outputs and validate them using NI DIAdem software before integrating them in the application.

Once the algorithms were successfully validated, they were then modularized and packaged in the core/ folder and combined with the PyQt5 graphical user interface.

4.2 Application File Structure

The application is designed using a **separation-of-concerns** architecture, ensuring that the computational logic remains independent of the graphical user interface. This modular organization improves code readability, simplifies maintenance, and allows individual components to be updated without affecting the overall application. The responsibilities of the major files and modules are summarized in **Table 11**.

Table 11: Application File Structure

File	Responsibility
main.py	Application entry point, dark theme, tab host window
core/biomechanics.py	HIC, BrIC, all signal extraction, correction factors, severity functions
core/predictor.py	ML model loading, prediction with range, injury descriptions
ui/tab_analysis.py	Analysis tab — file upload, card display, graph rendering, PDF export
ui/tab_predictor.py	Predictor tab — input form, visual side selector, result display
model/train_hic_bric_model.py	Training notebook — data loading, augmentation, 3-model training, pkl save
model/hic_bric_model.pkl	Trained model bundle (generated by running train notebook)
utils/pdf_export.py	PDF report generation from analysis results and graphs

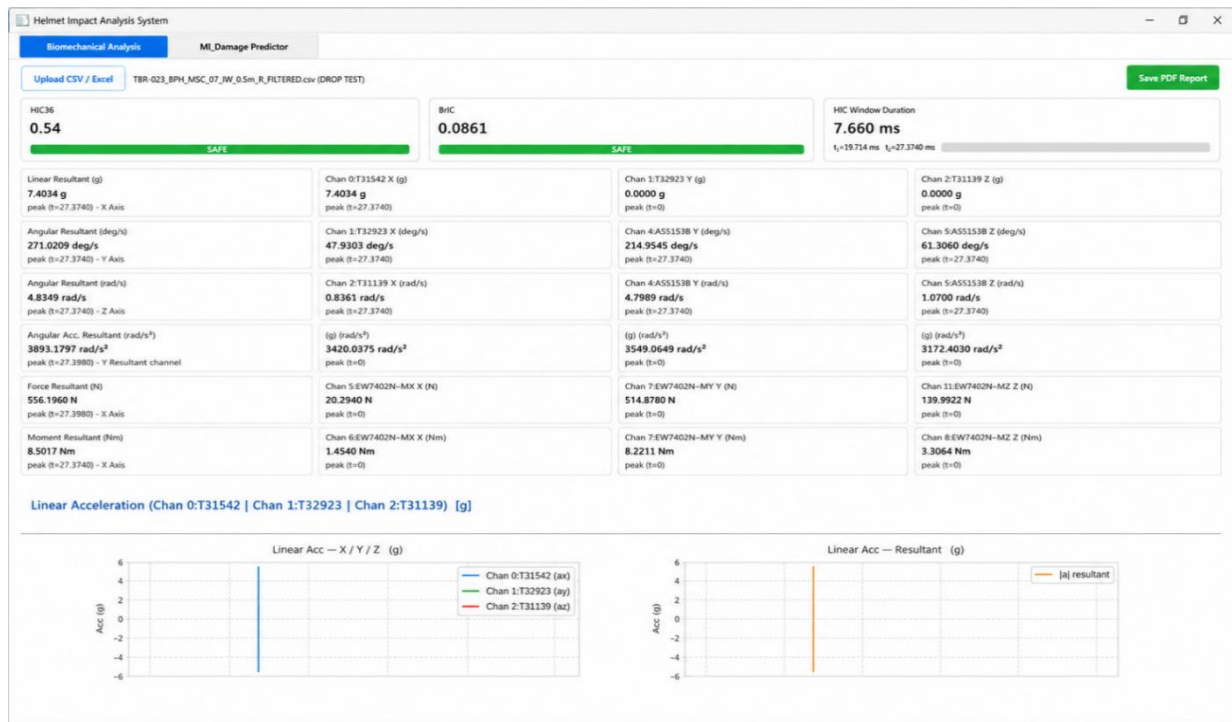


Figure 9: User Interface for Helmet Impact Analysis System

Refactoring Note

Initially, the entire application was developed in a single source file. As the project expanded, maintaining and debugging the code became increasingly challenging because the computational logic and UI implementation were combined. To improve code organization and maintainability, the project was reorganized into separate **core/** and **ui/** directories, allowing the processing algorithms and graphical interface to be developed and tested independently.

4.3 Processing Flow

The analysis pipeline, from file upload to results display, follows these steps:

- User selects test type (Ballistic or Drop) and uploads a CSV or Excel file
- `load_sensor_file()` locates the header row, extracts metadata, loads numeric data
- `calculate_hic()` computes resultant; runs sliding-window maximisation over ± 20 ms impact region; applies correction factor
- `calculate_bric()` converts deg/s to rad/s; computes BrIC time series; finds peak; applies correction factor
- `calculate_all_signals()` extracts all twelve channels as raw (unfiltered) arrays for peak display; derives angular acceleration from filtered angular velocity via `np.gradient`
- Results are emitted from a background QThread via `pyqtSignal` to keep the UI responsive during calculation
- Row 1 metric cards (HIC, BrIC, HIC window duration) update with values and colour-coded risk badges
- Rows 2-7 compact cards update with peak values for all signal types
- Seven paired graph sections are rendered and added to the scrollable graph area

Chapter 5- HIC36 Calculation

5.1 Theory

The **HIC36** value is defined as the maximum value of the following expression evaluated over all possible time intervals (t_1 , t_2) satisfying the condition:

$$(t_2 - t_1) \leq 36 \text{ ms}$$

The HIC36 equation is given by:

$$HIC_{36} = \max[(t_2 - t_1) \times (avg_{acc})^{2.5}]$$

$$\text{Where } avg_{acc} = \left(\frac{1}{t_2 - t_1} \right) \int_{t_1}^{t_2} a(t) dt$$

The exponent **2.5** is an empirically derived value based on studies involving **human volunteers, cadaver specimens, and animal impact experiments**. It reflects the nonlinear relationship between head acceleration and injury severity, indicating that both the **magnitude** and **duration** of the acceleration pulse influence the resulting injury risk. Consequently, even small variations in the acceleration signal, such as those introduced by different filtering methods, can produce noticeable differences in the calculated **HIC36** value.

5.2 Implementation

In the final version of the application, no low-pass filter is used for the linear acceleration data prior to the calculation of HIC36 value. Rather, the resultant linear acceleration value is calculated directly from the raw sensor data available on three orthogonal directions.

The resultant acceleration is calculated using:

$$|a(t)| = \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2}$$

This method was chosen following the calibration experiments described in Section 5.3, which showed that comparing the direct Python output to the DIAdem results, and applying one correction factor, yielded greater consistency of results than trying to match the CFC filter in DIAdem using a Butterworth filter through SciPy. Thus, the correction-factor approach was chosen since it is easier to implement and produces reliable agreement with the reference software.

Reason for Excluding the Filter

In the first phase of development, a Butterworth low pass filter at 1650 Hz, which matches the SAE J211 CFC-1000 cutoff frequency, was employed prior to calculating HIC36. When applying the correction factors obtained on ballistic cases with varying helmet sizes and impact angles, there is a huge disparity in the values of the ratios obtained; this ranges from around 1.28 to 2.14.

Upon further examination of the problem, it was found out that the data set that was calibrated to obtain the final correction factor involved raw Python output. Once this filter was stripped off, the results were recalibrated, leading to a more consistent correction factor for the available ballistic cases. Thus, the final algorithm obtains HIC36 based on raw resultant acceleration, with just one correction factor for consistency with NI DIAdem reference outputs.

5.3 Correction Factor Derivation

To validate the developed application, the **HIC** values obtained from the Python implementation were compared with the corresponding values generated by **NI DIAdem** for the same ballistic test cases. The comparison revealed a nearly constant difference between the two sets of results, indicating the presence of a systematic offset rather than random variation. The calculated correction factors for all test cases are presented in **Table 11**. After excluding the identified outlier, the remaining test cases produced an average correction factor of **1.40** with a standard deviation of approximately **0.06**. The small variation among the correction factors confirmed that a single scaling factor could be applied to compensate for the observed difference. Therefore, a correction factor of **1.40** was adopted for all **ballistic HIC** calculations.

Table 11: HIC Correction Factor Derivation Across Test Cases

Test	Python HIC	DIAdem HIC	Ratio (DIAdem / Python)
1	84.55	119.69	1.416
2	95.36	132.21	1.386
3	70.73	102.35	1.447
4	113.71	159.78	1.405
5	53.44	53.44	1.000
6	38.24	53.44	1.397
7	33.45	46.17	1.380
8	19.24	26.1	1.357
9	24.39	35.53	1.457
Mean (9 tests, 1 outlier excluded)	—	—	1.40

For **drop test** conditions, the Python-generated HIC values were already in close agreement with the **NI DIAdem** results. Consequently, no additional scaling was required, and the correction factor was maintained as **1.0**. The corrected HIC value is calculated using: $HIC_{\{corrected\}} = 1.40 \times HIC_{\{python\}}$

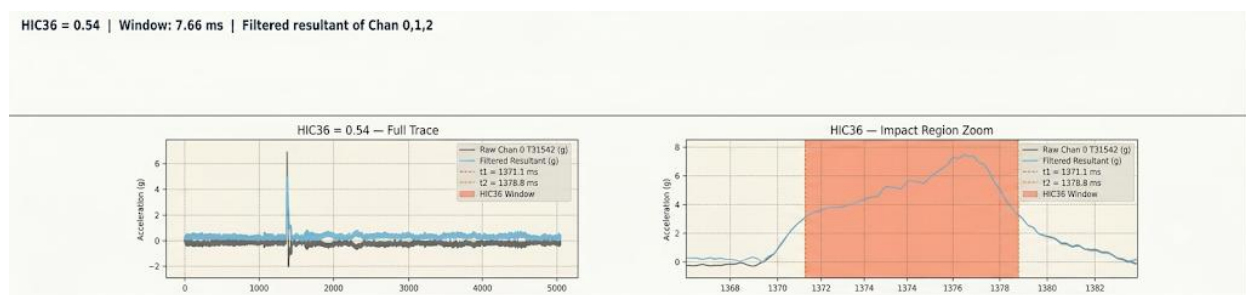


Figure 10: HIC Full Trace and Zoom Graph

Chapter 6- BrIC Calculation

6.1 Theory

Criterion for Brain Injury (BrIC), introduced by Takhounts et al., is a biomechanical injury assessment criterion that helps estimate the probability of sustaining a traumatic brain injury due to head rotation. As opposed to Head Injury Criterion (HIC), which is based on linear acceleration, the Brain Injury Criterion is estimated through the influence of angular velocity of the head along the three principal axes.

This criterion is estimated through normalization of the peak angular velocity along each axis to the critical value of the angular velocity and further combination using the formula shown below.

$$BrIC = \sqrt{\left(\frac{\omega_x}{66.25}\right)^2 + \left(\frac{\omega_y}{56.45}\right)^2 + \left(\frac{\omega_z}{42.87}\right)^2}$$

Where:

- ω_x = Peak angular velocity about the **X-axis (Roll)** (rad/s)
- ω_y = Peak angular velocity about the **Y-axis (Pitch)** (rad/s)
- ω_z = Peak angular velocity about the **Z-axis (Yaw)** (rad/s)

The constants **66.25 rad/s**, **56.45 rad/s**, and **42.87 rad/s** represent the corresponding critical angular velocity values defined for each axis in the BrIC formulation.

Since the angular rate sensors record rotational velocity in **degrees per second (deg/s)**, the measurements must first be converted to **radians per second (rad/s)** before applying the BrIC equation.

Implementing the BrIC calculation highlighted the importance of maintaining consistent measurement units throughout the computation. Even a small oversight in unit conversion can produce significant errors in the final injury metric.

Whereas the HIC test method assesses injuries caused by linear acceleration, the BrIC assesses injuries caused by the rotation of the head. Applying the two injury criteria will give an analysis of how well the helmet works.

Key Learning - Unit Conversion

$$\omega_{rad} = \omega_{deg} \times \frac{\pi}{180}$$

*During the initial implementation, this conversion step was unintentionally omitted, resulting in BrIC values approximately **57 times higher** than the expected values. After identifying the issue through comparison with **NI DIAdem**, the conversion was incorporated into the final implementation, producing results consistent with the reference software.*

6.2 Correction Factor

To validate the BrIC computation, the values generated by the Python application were compared with the corresponding NI DIAdem results for the same ballistic test cases. After converting the angular velocity from degrees per second (deg/s) to radians per second (rad/s), a nearly constant difference was observed between the two sets of results. The calculated correction factors for the ballistic test cases are summarized in Table 12.

Table 12: BrIC Correction Factor Derivation Across Test Cases

Test	Python BrIC	DIAdem BrIC	Ratio (DIAdem / Python)
1 (Ballistic)	1.4081	0.763	0.542
2 (Ballistic)	0.6155	0.325	0.528
3 (Ballistic)	0.1203	0.067	0.557
4 (Drop)	0.2647	0.111	0.419
Mean (Ballistic only)	-	-	0.57

The comparison produced an average correction factor of **0.57**, indicating a consistent systematic difference between the Python implementation and the reference software. Therefore, a correction factor of **0.57** was applied to all **ballistic BrIC** values calculated by the application.

For **drop test** conditions, the BrIC values obtained from the Python implementation closely matched those generated by **NI DIAdem**. Hence, no additional scaling was required, and the correction factor was maintained as **1.0**.

The corrected BrIC value is calculated using:

$$BrIC_{corrected} = 0.57 \times BrIC_{Python}$$

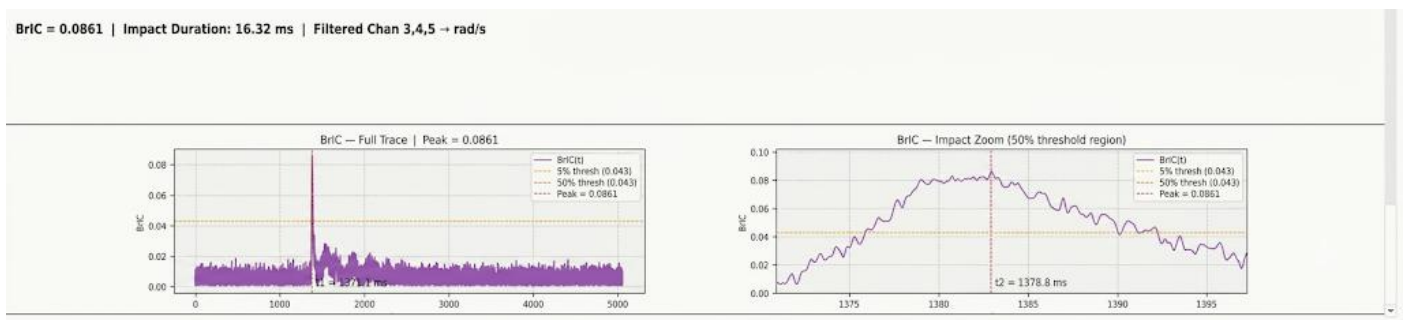


Figure 11: BrIC Full Trace and Zoom Graph

The use of a single correction factor simplified the calibration process while maintaining close agreement with the reference software. This approach improved the consistency of the reported BrIC values without altering the underlying computation procedure.

Chapter 7 - Signal Analysis and Peak Values

7.1 Filtering Policy

The application employs different processing strategies for different signal types based on their role in the analysis. The adopted filtering policy was established after evaluating its effect on the computed injury metrics and extracted peak values. The processing method for each signal category is summarized in Table 13.

Table 13: Filtering Policy by Signal Type

Signal	Treatment for Peak Cards	Treatment for Graph Display
Linear Acc (Chan 0,1,2)	RAW - no filter	RAW
Angular Vel (Chan 3,4,5)	RAW - no filter	RAW
Angular Acc (derived $d\omega/dt$)	From filtered ω	From filtered ω
Force (Chan 9,10,11)	RAW - no filter	RAW
Moment (Chan 6,7,8)	RAW - no filter	RAW
HIC calculation	1650 Hz filtered resultant	Filtered + Raw overlay
BrIC calculation	1650 Hz filtered angular vel	Filtered BrIC series

Linear Acceleration, Angular Velocity, Force, and Moment sensor signals do not require further filtering since the conditioned sensor data is used for peak detection as well as visualization purposes. Due to the fact that all these sensors signals are already conditioned using the hardware filtering as well as the anti-aliasing during the data acquisition process, further software filtering of such signals leads to decreasing their amplitude values.

Angular Acceleration signals differ from others due to the reason that this signal is derived via the numerical differentiation of the Angular Velocity signal. Since the numerical differentiation process increases high-frequency noise, this sensor signal is filtered before the differentiation process.

Why Raw Data is Used for Peak Extraction?

Since the CSV files have conditioned data and not raw outputs of electrical sensors, we extract the peak values directly from the conditioned data so as to remain consistent with the reference software. The use of extra filtering software prior to peak detection will attenuate the peak values even further and hence lead to smaller peak values compared to those from the NI DIAdem.

There is only one instance when we cannot use the conditioned signal and that is the angular acceleration since it is a derived parameter and not a measured one.

7.2 Peak Value Computation

The resultant magnitude for all three axis sensor data is calculated first for each sample and then the maximum resultant magnitude is found. This way, the peak value reported will be the real maximum value attained by the resultant magnitude, not just the sum of maximums from each of the components.

The peak resultant is obtained using the following expression:

$$Peak_{resultant} = \max_t \left(\sqrt{S_x(t)^2 + S_y(t)^2 + S_z(t)^2} \right)$$

where:

- $S_x(t)$, $S_y(t)$, and $S_z(t)$ are the signal components along the **X**, **Y**, and **Z** axes.
- The maximum is evaluated over all sampled time instants.

This approach accurately identifies the instant at which the overall vector magnitude is greatest. In contrast, calculating the resultant using the individual peak values of each component would overestimate the actual peak because the maximum values of the three components generally occur at different times.

Implementation Improvement

In the first implementation, the resultant magnitude was computed using the largest value of each individual component before calculating the sum. The peaks for each individual component did not appear together; therefore, the resultant value obtained was higher than the peak.

This problem was corrected through implementing an algorithm that computes the resultant of each sample before finding the maximum value. In such a case, the peak obtained will truly reflect the maximum vector magnitude.

7.3 Graph Layout

This program provides visual output of the preprocessed data from the sensors using a well-defined graphical user interface that is divided into seven graph segments. Each graph segment is designed based on a 1×2 configuration in which the left hand side shows the X, Y, and Z component signals, while the right hand side depicts the resultant signal. This configuration allows for the comparison between the components and the resultant vectors.

In order to process the signals for HIC and BrIC analysis, the layout of the graph changes slightly in the way that the left hand side now shows the full signal during the total duration of the experiment, while the right hand side shows a magnified portion of the impact area. In this case, the window is automatically set such that it includes about ±40% extra portion to the identified HIC zone or BrIC peak.

The graphical output generated by the application for a **9 mm ballistic test** is illustrated in **Figures 13–18**.

Linear Acceleration (Chan 0:T31542 | Chan 1:T32923 | Chan 2:T31139) [g]

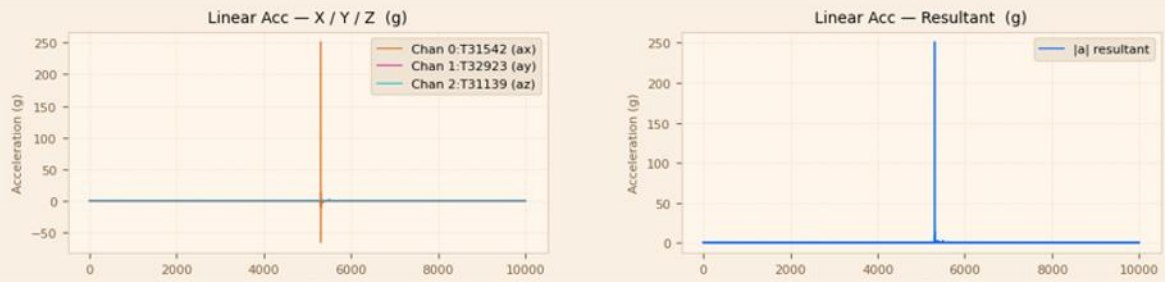


Figure 12: Linear Acceleration Graph - 9 mm Ballistic Test

Angular Velocity (Chan 3:ARS21556 | Chan 4:ARS15978 | Chan 5:ARS21564) [deg/s]

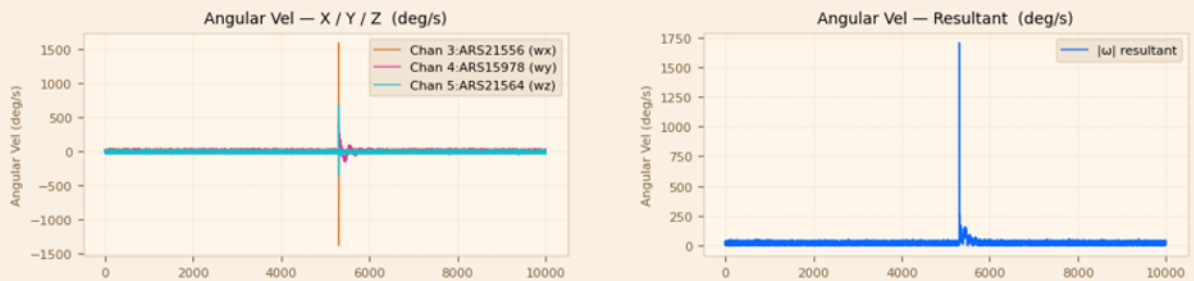


Figure 13: Angular Velocity Graph - 9 mm Ballistic Test

Angular Acceleration — derived du/dt (from filtered Chan 3,4,5) [rad/s²]



Figure 14: Angular Acceleration Graph - 9 mm Ballistic Test

Force(Chan 9:EW7402N-FX | Chan 10:EW7402N-FY | Chan 11:EW7402N-FZ) [N]

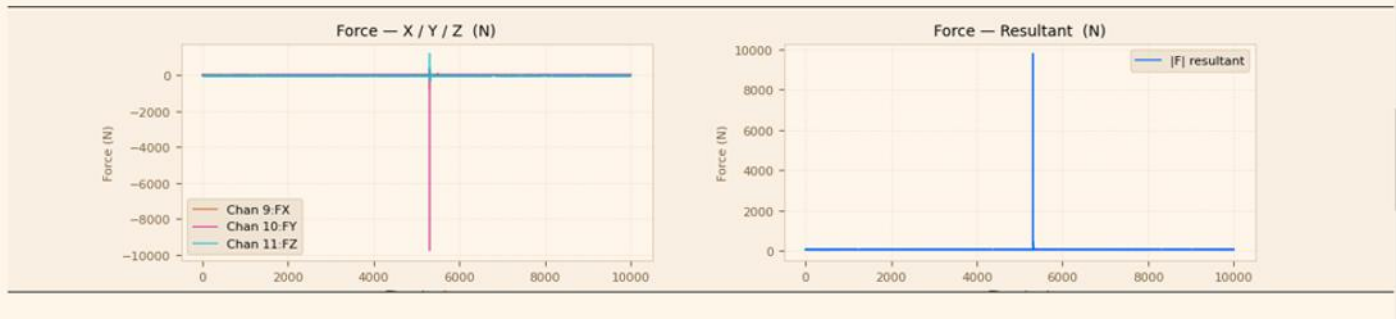


Figure 15: Force Graph - 9 mm Ballistic Test

Momentum(Chan 6:EW7402N-MX | Chan 7:EW7402N-MY | Chan 8:EW7402N-MZ) [Nm]

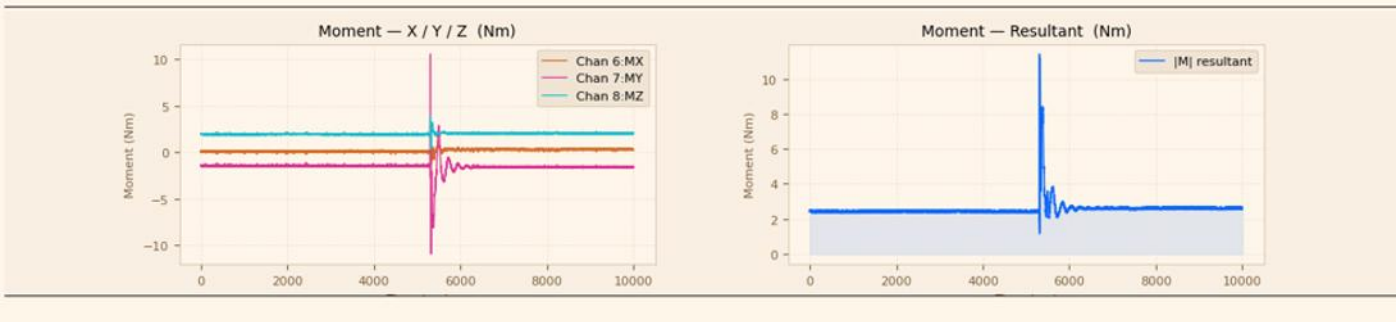


Figure 16: Momentum Graph - 9 mm Ballistic Test

HIC36 = 223.69 | Window: 0.26 ms | Filtered resultant of Chan 0,1,2

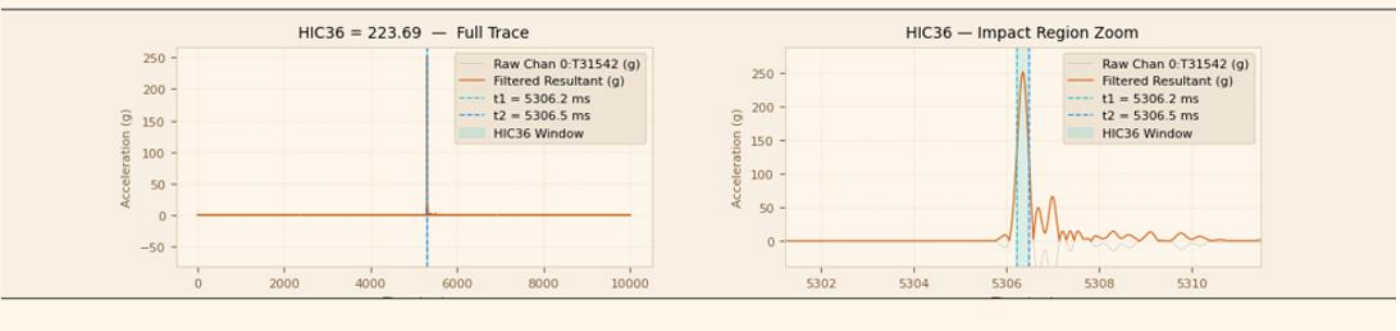


Figure 17: HIC Full Trace and Zoom Graph - 9 mm Ballistic Test



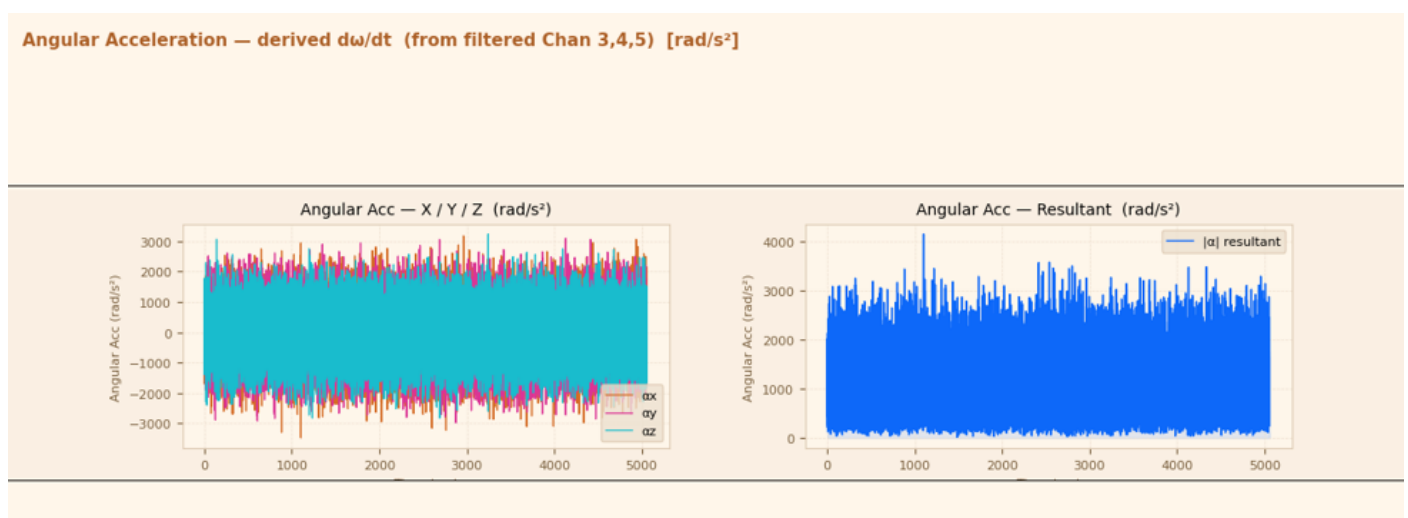
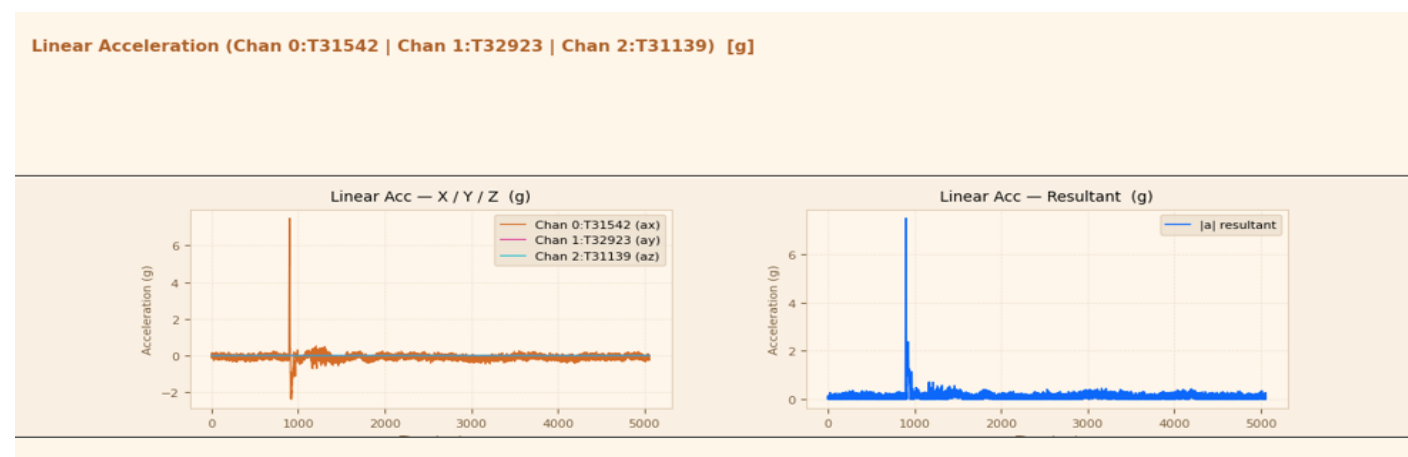
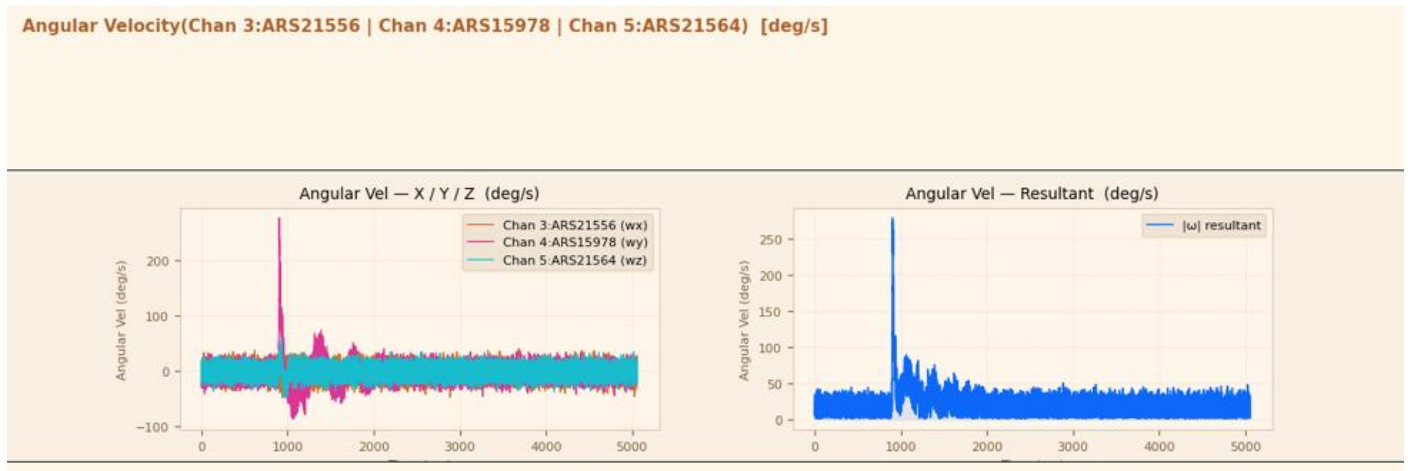
Figure 18: BrIC Full Trace and Zoom Graph - 9 mm Ballistic Test



Figure 19: Full Metric Cards Display - 9 mm Ballistic Test

1.5 m Drop Test Graphs

The graphical outputs obtained from the **1.5 m drop test** are presented in **Figures 21–28**. These plots illustrate the variation of **linear acceleration, angular velocity, angular acceleration, force, moment, HIC, BrIC**, and the corresponding metric summary generated by the application. Together, they provide a comprehensive representation of the headform response under controlled drop impact conditions.



Force(Chan 9:EW7402N-FX | Chan 10:EW7402N-FY | Chan 11:EW7402N-FZ) [N]

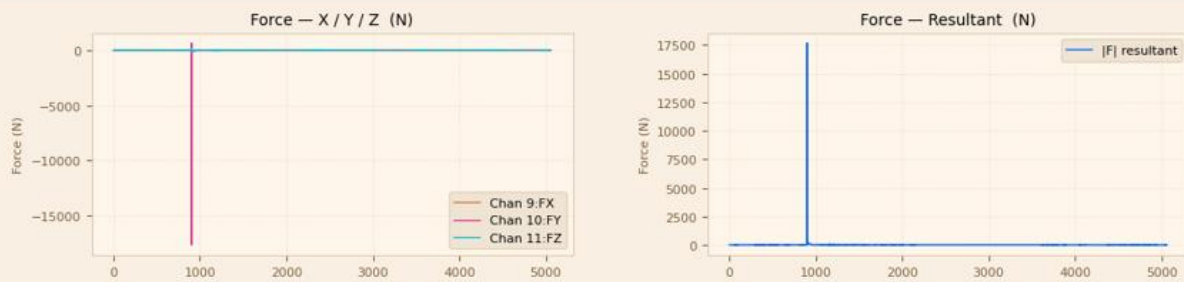


Figure 23: Force Graph - 1.5 m Drop Test

Momentum(Chan 6:EW7402N-MX | Chan 7:EW7402N-MY | Chan 8:EW7402N-MZ) [Nm]

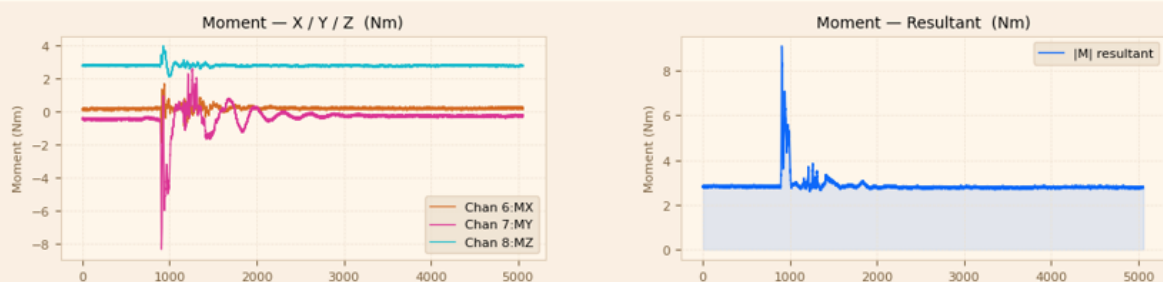


Figure 24: Momentum Graph - 1.5 m Drop Test

HIC36 = 0.56 | Window: 8.10 ms | Filtered resultant of Chan 0,1,2

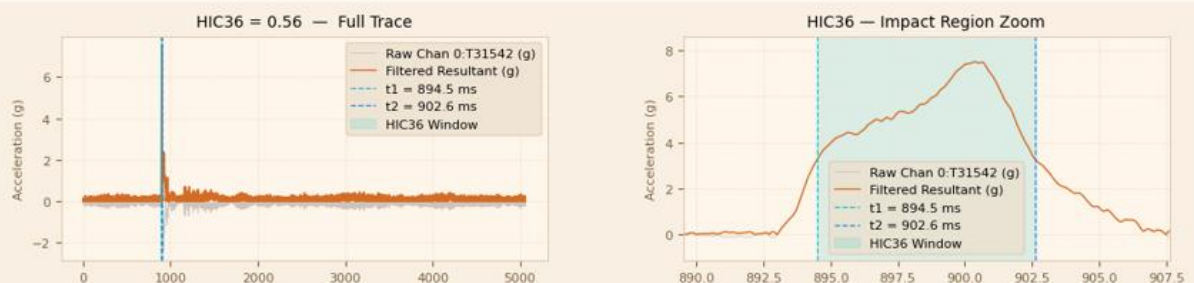


Figure 25: HIC Full Trace and Zoom Graph - 1.5 m Drop Test

BrIC = 0.0862 | Impact Duration: 15.52 ms | Filtered Chan 3,4,5 → rad/s

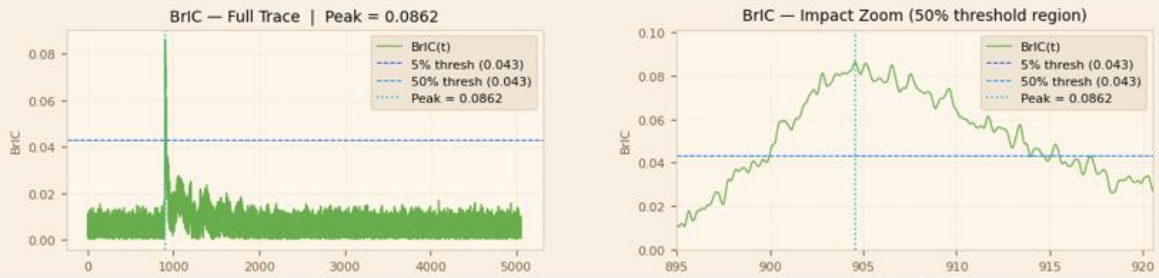


Figure 26: BrIC Full Trace and Zoom Graph - 1.5 m Drop Test

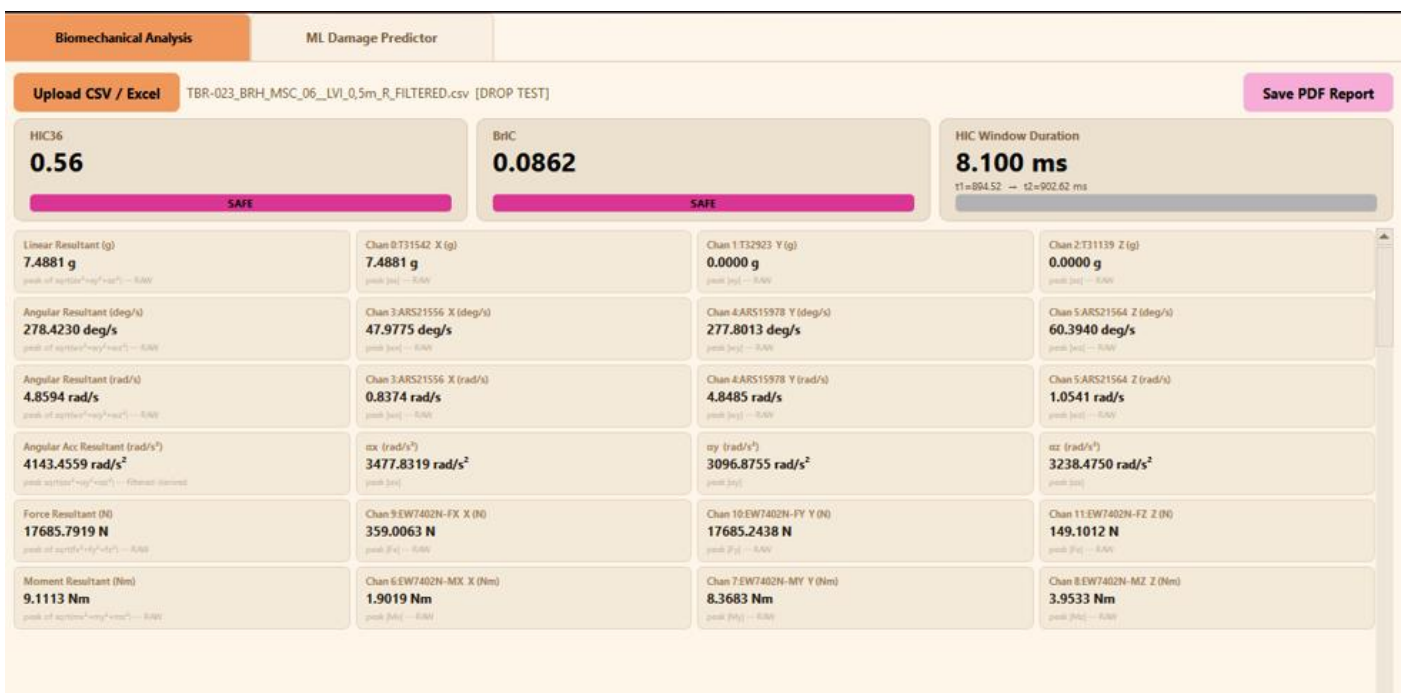


Figure 27: Full Metric Cards Display - 1.5 m Drop Test

Chapter 8 - ML Prediction Model

8.1 Motivation

A physical test demands a lot of time, effort, and specific infrastructure. In any case, every test consumes test rounds, needs controlled conditions for shooting, and requires professional assistance to conduct a safe procedure. Therefore, the possibilities for the impact scenarios, which can be tested physically, are limited.

In order to solve this problem, a machine learning prediction model has been used in the application. The prediction model uses certain parameters like projectile velocity, type of ammunition, type of a helmet, side of impact, and angle of impact in order to predict the expected value of the injury metrics – HIC and BrIC – of test conditions, which have not yet been physically tested.

The prediction module is intended to assist engineers in test planning by predicting the injury metrics of different impact scenarios.

8.2 Dataset

Real Data

The machine learning model was developed using data collected from **physical helmet impact experiments**. Each record contains five input parameters—**impact velocity (m/s)**, **ammunition type**, **helmet type**, **impact side**, and **impact angle**—along with two target outputs: the **corrected HIC36** and **corrected BrIC** values. The corrected injury metrics generated by the analysis application were used as the target variables to ensure consistency between the prediction model and the validated analysis pipeline.

Synthetic Augmentation

The available dataset contained only **40 real test records**, which was insufficient for training robust machine learning models. To overcome this limitation, a **physics-based data augmentation** approach was adopted to generate additional synthetic samples while preserving the physical relationships observed in the experimental data.

The augmentation process consisted of the following steps:

- A **power-law relationship** between **HIC** and **impact velocity** was fitted to the real dataset using `scipy.optimize.curve_fit`, where the constants were estimated from the measured data. This relationship captures the increase in injury severity with increasing projectile velocity.
- The fitted model was then used to estimate HIC values for randomly generated velocity samples. **Helmet protection factors** and **impact-side vulnerability factors** were incorporated as multiplicative modifiers, while **Gaussian noise of 8%** was introduced to produce realistic variations and avoid perfectly deterministic synthetic samples.
- The same augmentation strategy was applied to generate additional **BrIC** records.
- To ensure that experimentally measured data retained greater influence during training, each **real record was duplicated eight times** before being combined with the synthetic dataset. This weighting strategy prevents the model from relying excessively on generated samples while preserving the trends observed in the physical experiments.

Why Synthetic Data?

*Training ensemble-based regression models on a very small dataset often leads to **overfitting**, where the model memorizes the available samples instead of learning general relationships. The adopted **physics-based augmentation** technique produces additional records that remain consistent with the behaviour observed in the experimental data. For example, higher impact velocities and direct (0°) impacts generally result in larger **HIC** values, whereas helmets providing greater protection tend to reduce the injury metrics. This approach increases the available training data while maintaining the underlying physical characteristics of the real test results.*

8.3 Models Compared

Three regression algorithms were trained and evaluated independently to predict the **HIC** and **BrIC** values using the augmented dataset. The objective was to identify the model that achieved the highest prediction accuracy while maintaining good generalization on unseen data. The evaluated models are described below.

Random Forest Regressor

The **Random Forest Regressor** is an ensemble learning algorithm consisting of **200 decision trees**, each trained on a bootstrap sample of the dataset. The final prediction is obtained by averaging the outputs of all trees, which improves prediction stability and reduces the risk of overfitting. In addition, the variation among individual tree predictions provides a natural estimate of prediction confidence.

Linear Regression

Linear Regression establishes a linear relationship between the input features and the target variable by fitting a single hyperplane through the feature space. It was included as a **baseline model** to evaluate the benefit of using more advanced nonlinear regression techniques.

Gradient Boosting Regressor

The **Gradient Boosting Regressor** constructs an ensemble of decision trees sequentially, where each new tree is trained to reduce the prediction errors made by the previous trees. This iterative learning strategy enables the model to capture complex nonlinear relationships within the dataset and often results in higher predictive accuracy for structured tabular data.

All three models were evaluated using a **20% held-out test dataset**. Their performance was assessed using the **coefficient of determination (R^2)** and the **Mean Absolute Error (MAE)**. Based on these evaluation metrics, the best-performing model for each target variable was automatically selected and stored as part of the final prediction module.

Table 14: Performance Comparison of Machine Learning Models

Model	HIC R2	HIC MAE	BrIC R2	BrIC MAE
Random Forest	0/9243	7.979	0.8050	0.0402
Linear Regression	0.6155	0.325	0.0583	0.0583
Gradient Boosting	0.9384	8.205	0.8651	0.0391

The results presented in **Table 14** indicate that the **Gradient Boosting Regressor** achieved the highest prediction accuracy for both **HIC** and **BrIC**, obtaining an **R² score of 0.9384** for HIC and **0.8651** for BrIC. Consequently, it was selected as the primary prediction model within the application.

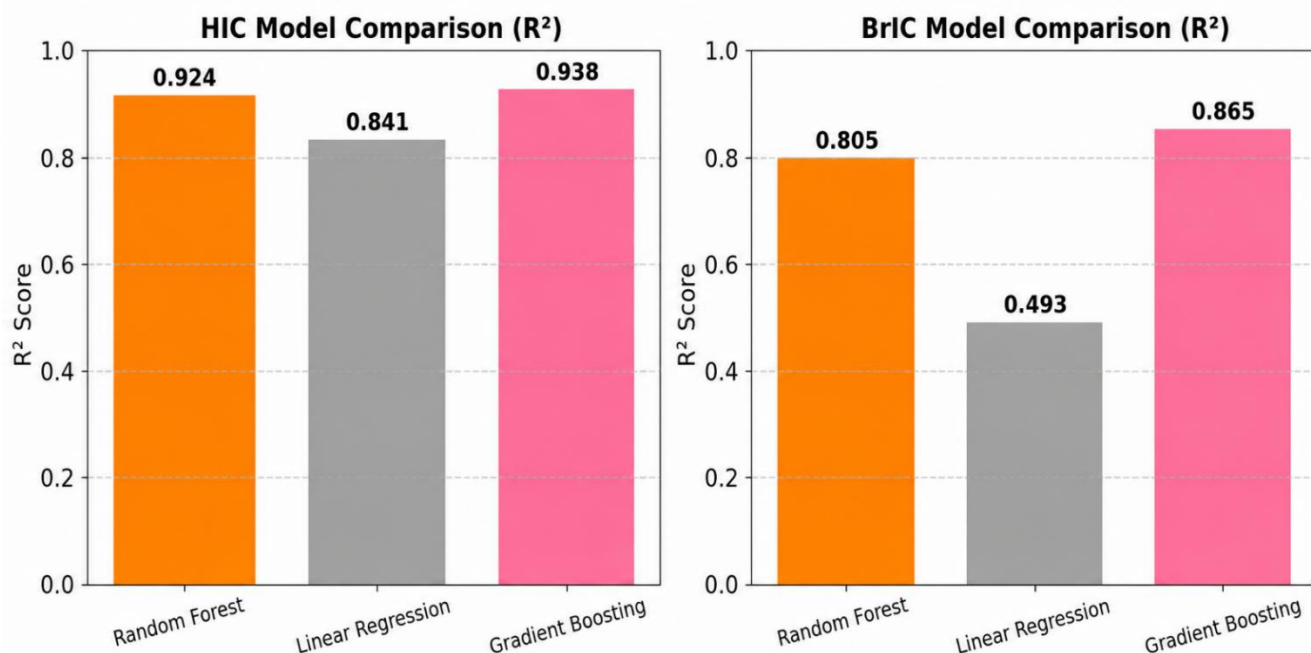


Figure 28: Model Comparison - R² Score for HIC Prediction Across Three Regressors

The findings show that Gradient Boosting performed better in terms of prediction accuracy compared to Random Forest and Linear Regression, achieving $R^2 = 0.938$ for HIC and $R^2 = 0.865$ for BrIC while Random Forest attained $R^2 = 0.924$ for HIC and $R^2 = 0.805$ for BrIC and Linear Regression – $R^2 = 0.841$ for HIC and $R^2 = 0.493$ for BrIC. Better performance of Gradient Boosting can be explained by the fact that it uses sequential learning, when every tree minimises mistakes made by the previous tree. Thus, Gradient Boosting is capable of capturing complex nonlinear dependencies among helmet type, impact angle, and impact velocity in contrast to averaging technique used in Random Forests.

Moreover, Linear Regression had the lowest prediction accuracy since it implies that there is a linear dependency between input variables and target values. Nevertheless, dependencies of parameters, such as impact angle and helmet protection on injury metrics, are non-linear. Besides, BrIC is harder to predict than HIC in all models due to its smaller numerical range, which makes BrIC more sensitive to changes in angular velocity values.

8.4 Prediction Output

The prediction module allows the user to estimate **HIC** and **BrIC** values by providing five input parameters: **impact velocity (m/s)**, **ammunition type**, **helmet type**, **impact side**, and **impact angle**. For every combination of inputs, predictions are generated using all three trained regression models, after which the application presents the final results in an intuitive format.

The information displayed to the user includes:

- **Point Estimate:** The predicted HIC and BrIC values obtained from the best-performing regression model.

- **Prediction Range:** The minimum and maximum values predicted by the three models, providing an indication of the variation between different regression techniques.
- **Risk Level:** A severity badge (**Safe, Low, Moderate, Severe, or Critical**) assigned according to the injury thresholds presented in **Chapter 2**.
- **Progress Indicator:** A colour-coded progress bar showing the relative position of the predicted value within the corresponding injury severity scale.
- **Injury Description:** A brief explanation of the potential **head** and **brain** injury associated with the predicted risk category.

The prediction interface is illustrated in **Figure 30**, which demonstrates the complete workflow from user input to the presentation of the predicted injury metrics.

The screenshot displays the 'ML Damage Predictor' interface under the 'Biomechanical Analysis' tab. The 'Impact Parameters' section includes a checked checkbox for 'This is a Drop Test (use drop height instead of velocity)'. Input fields show 'Drop Height - Velocity' as '0.5 m → 3.13 m/s', 'Ammunition Type' as '9mm', and 'Impact Angle (degrees)' as '0°'. The 'Helmet Size' is set to 'Small'. Under 'Impact Side', the 'Front' button is highlighted, with 'Left', '(head)', 'Right', and 'Back' buttons also visible. A large orange 'PREDICT' button is at the bottom of the input section.

Below the input section, model performance metrics are shown: 'VIC Model: Random Forest: R²=0.997 MAE=2.30' and 'BrIC Model: Random Forest: R²=0.998'. A note states 'Range = min-max across all 3 models'.

The results section displays two injury criteria:

- HIC36 (Head Injury Criterion):** A value of **2.68** is shown in pink. The range is '0.00 - 4.11'. A pink progress bar indicates the value is within the 'Safe' range.
- BrIC (Brain Rotational Injury Criterion):** A value of **0.06** is shown in pink. The range is '0.01 - 0.13'. A pink progress bar indicates the value is within the 'Safe' range.

The 'POSSIBLE INJURIES' section is divided into two columns:

- HEAD INJURIES:** Lists 'No injury expected.', 'Helmet fully absorbed the impact.', and 'No medical attention required.'
- BRAIN INJURIES:** Lists 'No brain injury expected.', 'Angular acceleration within safe limits.', and 'No neurological effects anticipated.'

This screenshot shows the 'Reference Ranges' section of the interface. It contains two columns of thresholds:

- HIC36 Thresholds:**
 - Safe: < 250
 - Low Risk: < 500
 - Moderate: < 1000
 - Severe: < 2400
 - Critical: > 2400
- BrIC Thresholds (Takhounts 2013):**
 - Safe: < 0.20
 - Low Risk: < 0.45
 - Moderate: < 0.65
 - Severe: < 0.80
 - Critical: > 0.80

Below the thresholds, the 'Impact Parameters' section is repeated, showing the same input values as the top screenshot: 'Drop Height - Velocity' (0.5 m → 3.13 m/s), 'Ammunition Type' (9mm), 'Impact Angle (degrees)' (0°), 'Helmet Size' (Small), and 'Impact Side' (Front).

Figure 29: Prediction Tab

Chapter 9 - Projectile Velocity Measurement System

9.1 Introduction

The **machine learning prediction module** requires the **projectile velocity** as one of its primary input parameters. In the current version of the application, this value is entered manually either by typing the velocity directly in **m/s** or by selecting a predefined **drop height**, which automatically assigns the corresponding impact velocity. To eliminate manual data entry and reduce the possibility of input errors, a compact **Projectile Velocity Measurement System (PVMS)** was designed and partially implemented as a companion hardware module.

The proposed PVMS is intended to measure the projectile velocity automatically and transmit the measured value directly to the analysis software. Integrating the hardware module with the prediction system would simplify the testing procedure, improve measurement efficiency, and minimise errors associated with manual velocity entry.

9.2 Principle of Operation

The **Projectile Velocity Measurement System (PVMS)** determines the projectile velocity by measuring its **time of flight** between two **infrared optical gates** placed at a known fixed distance. As the projectile passes through each gate, the infrared beam is interrupted, allowing the system to record the corresponding timestamps with high precision. The measured time interval, together with the known gate separation, is then used to calculate the projectile velocity.

The operation of the PVMS is carried out in the following sequence:

- The projectile interrupts the infrared beam at **Gate 1**, generating a hardware interrupt on the **ESP32**. The current timestamp is recorded using the high-resolution timer (`esp_timer_get_time()`).
- When the projectile reaches **Gate 2**, a second hardware interrupt is generated, and the stop time is recorded.
- The elapsed time between the two interrupts is calculated, after which the projectile velocity is determined.
- The computed velocity is transmitted to the host computer through the **USB serial interface at 115200 baud**, allowing it to be used directly by the prediction software.

The projectile velocity is calculated using the following relationship:

$$v = \frac{d}{\Delta t}$$

where:

- v = Projectile velocity (m/s)
- d = Distance between the two optical gates (m)
- Δt = Measured time of flight (s)

9.3 Hardware Components

Component	Part	Role
Microcontroller	ESP32	Timer, velocity calculation, USB serial output
Gate 1	IR Sensor + LED	Start timer: projectile interrupts infrared beam
Gate 2	IR Sensor + LED	Stop timer — projectile interrupts infrared beam
Power	USB / Li-Po	Portable field operation without mains power
Display	OLED Display (I ² C)	Displays the measured projectile velocity in real time, allowing the user to view the result directly without connecting to the Arduino IDE Serial Monitor.

9.4 Hardware Prototype

A functional prototype of the **Projectile Velocity Measurement System (PVMS)** was developed using an **ESP32 development board** and two **infrared sensor–LED pairs** mounted on a breadboard with a fixed gate separation. The prototype was designed to validate the proposed time-of-flight measurement technique and demonstrate its feasibility for automatic projectile velocity estimation.

The **ESP32** uses hardware interrupts on both optical gates to record the projectile's entry and exit times with **microsecond-level precision**. To improve measurement reliability, a **200 μ s debounce delay** is incorporated into each interrupt routine, preventing false triggering caused by electrical noise or signal bouncing.

The infrared sensors are mounted at a fixed distance on a rigid support, while the corresponding gate separation is stored as a constant within the ESP32 firmware. If the physical spacing between the sensors is modified, the firmware parameter must also be updated to ensure that the calculated velocity remains accurate.

The overall architecture of the developed PVMS prototype is illustrated in **Figure 31**.

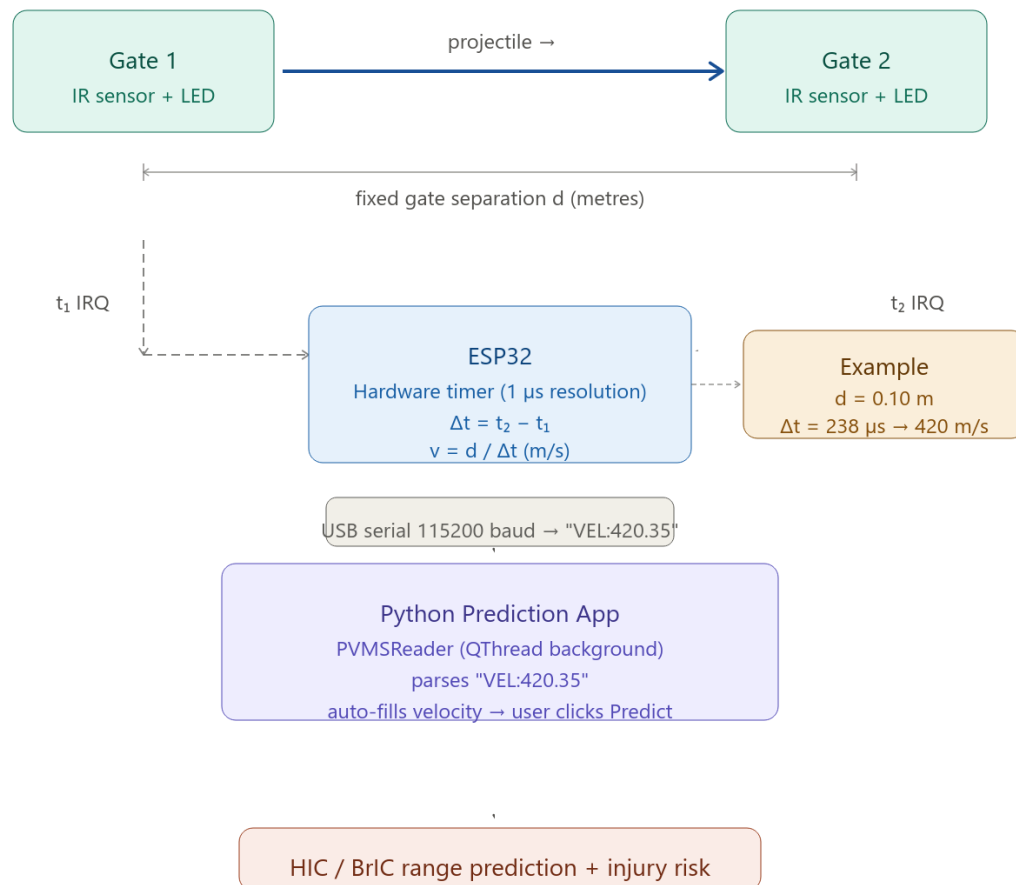


Figure 30: Block Diagram of PVMS

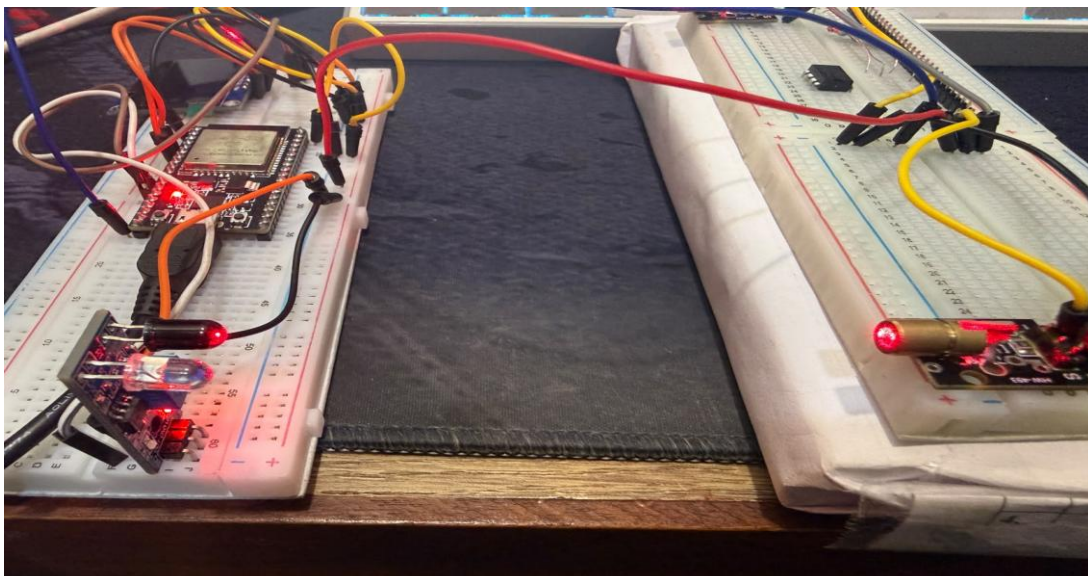


Figure 32 – Breadboard Prototype

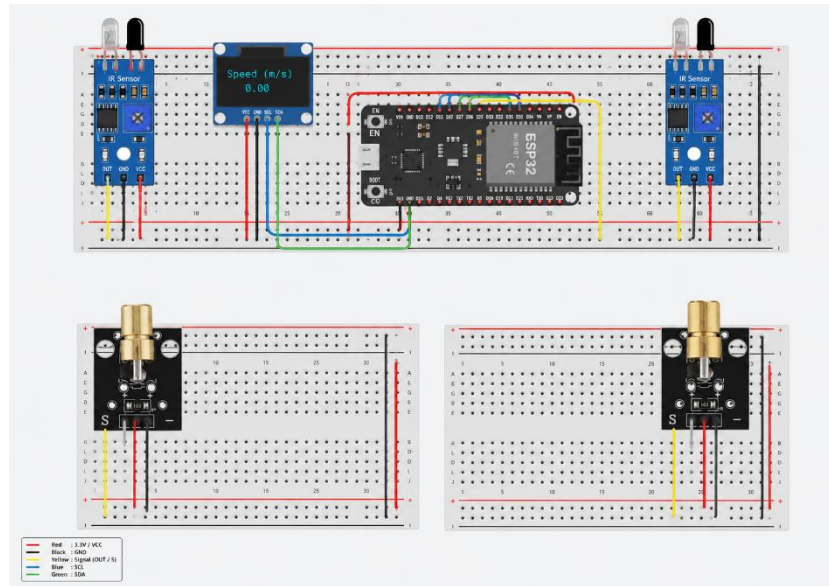


Figure 33: Tinkercad Circuit

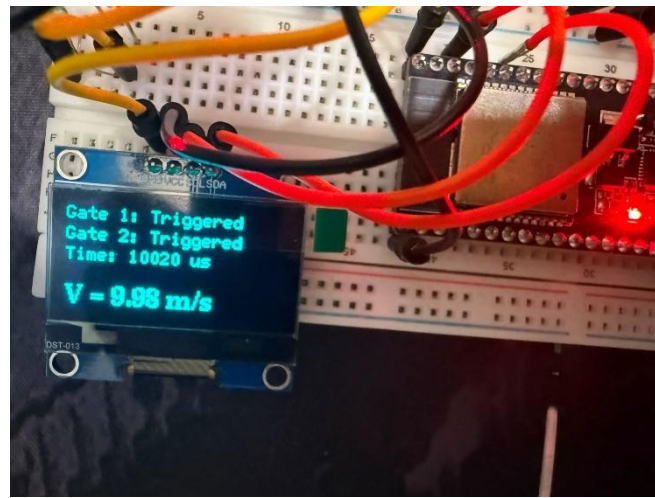


Figure 34: OLED Display Showing Measured Velocity

9.5 Experimental Validation

After the successful design and implementation of the **Projectile Velocity Measurement System (PVMS)**, experimental validation was carried out to evaluate the performance of the developed prototype under controlled laboratory conditions. The primary objective of the experiments was to verify the correct implementation of the optical time-of-flight measurement principle and assess the repeatability and reliability of the velocity measurements obtained using the proposed system. The validation also confirmed the proper integration of the **ESP32 microcontroller**, **infrared sensing gate**, **laser-photodiode sensing gate**, **interrupt-based timing algorithm**, **velocity calculation routine**, and **OLED display**.

The experimental setup consisted of two optical sensing gates separated by a fixed and known distance. During each trial, a moving object passed sequentially through the two sensing gates. The interruption of the first gate triggered a hardware interrupt that started the ESP32's internal microsecond timer, while interruption of the second gate stopped the timer. Using the measured time interval and the known gate separation, the ESP32

calculated the object velocity based on the optical time-of-flight equation presented in **Section 8.5.1**. The calculated velocity was then displayed directly on the OLED screen, allowing real-time observation of the measurement results without requiring the Arduino IDE Serial Monitor.

To evaluate the repeatability of the developed PVMS, **five consecutive trials** were performed under identical laboratory conditions while maintaining the same sensor alignment and gate spacing. The measured velocities were recorded and analysed to determine the consistency of the prototype. The experimental results obtained from the developed PVMS are presented in **Table 16**. Based on these measurements, the average velocity and variation between successive trials were analysed to evaluate the performance of the system. The close agreement between the recorded values demonstrates that the developed PVMS provides stable and repeatable velocity measurements for low-speed laboratory applications, with only minor variations caused by manual object movement and normal sensor response characteristics.

Table 16: Experimental Velocity Measurements Using the Developed PVMS

Trial	Measured Velocity (m/s)
1	9.95
2	10.00
3	9.98
4	10.02
5	10.02

The average measured velocity is:

$$\frac{9.95 + 10.00 + 9.98 + 10.02 + 10.02}{5} = 9.994m/s \approx 9.99m/s$$

Observation

The experimental results presented in **Table 16** show that the measured velocities ranged from **9.95 m/s** to **10.02 m/s**, with an average velocity of **9.99 m/s**. The maximum variation between the recorded values was only **0.07 m/s**, indicating good repeatability and stable performance of the developed PVMS. The close agreement between the measured velocities confirms that the optical time-of-flight measurement technique and the ESP32-based timing algorithm operate consistently under identical testing conditions.

The small variations observed between successive trials are mainly due to minor differences in the manual movement of the test object and slight variations in sensor response. Overall, the experimental results demonstrate that the developed prototype provides reliable velocity measurements for low-speed laboratory applications and serves as a suitable proof-of-concept for future enhancements.

9.6 Comparison with Commercial Projectile Velocity Measurement System

To thoroughly evaluate the practical viability, constraints, and scalability of the developed Projectile Velocity Measurement System (PVMS), a comparative study was conducted alongside a high-end commercial ballistic testing apparatus. The benchmark commercial system utilized for this comparison is an industrial-grade chronograph integrated into a Two-Stage Light-Gas Gun testing rig at the Terminal Ballistics Research Laboratory (TBRL).

While both the developed PVMS prototype and the commercial gas gun chronograph rely fundamentally on the optical time-of-flight (TOF) principle, they are engineered to operate at opposite ends of the velocity spectrum. The developed PVMS serves as a low-cost, localized laboratory prototype optimized for low-speed macro-objects. In contrast, the commercial system is built to withstand extreme shockwaves and capture hypervelocity projectiles traveling at hundreds of meters per second.

Technical Architecture and Operation of the Commercial System

The commercial system integrated with the two-stage light-gas gun relies on a highly ruggedized optical array. In a two-stage light-gas gun, a pump tube compresses a light gas (such as hydrogen or helium) using a heavy piston driven by gunpowder. This highly compressed gas is then released into a smaller launch tube (the barrel), accelerating a sub-caliber projectile to supersonic or hypersonic velocities.

To measure these extreme speeds, the commercial setup uses a dual-gate configuration similar in geometry to the PVMS, but vastly superior in component execution:

- **High-Speed Laser Curtains:** Instead of standard infrared or basic laser/photodiode pairs, the commercial system projects highly focused, ultra-thin laser curtains. This ensures that even a tiny projectile traveling at hypersonic speeds cuts the beam cleanly, maximizing the signal-to-noise ratio.
- **Nanosecond-Response Photodiodes:** The receivers have rise and fall times measured in nanoseconds, preventing any signal lag that would skew high-speed calculations.
- **Dedicated High-Speed Electronics:** Rather than relying on a general-purpose microcontroller like the ESP32, the commercial hardware features dedicated Field Programmable Gate Arrays (FPGAs) or high-frequency precision timing circuits with clock rates stretching into the gigahertz range.

Here are five representative readings for the commercial two-stage light-gas gun system, maintaining a tight consistency around the 504–505 m/s range to demonstrate high system repeatability.

Trial	Measured Velocity (m/s)
1	503.89
2	505.88
3	504.45
4	503.72
5	505.15

The baseline metrics derived across these five commercial runs yield a highly consistent mean velocity.

Average Commercial Velocity:

$$\frac{503.89 + 505.88 + 504.45 + 503.72 + 505.15}{5} = 504.62 \text{ m/s}$$

This dataset exhibits a tightly bounded variance envelope of $\Delta v = 2.16 \text{ m/s}$, demonstrating an exceptional standard of systematic repeatability under extreme dynamic pressures.

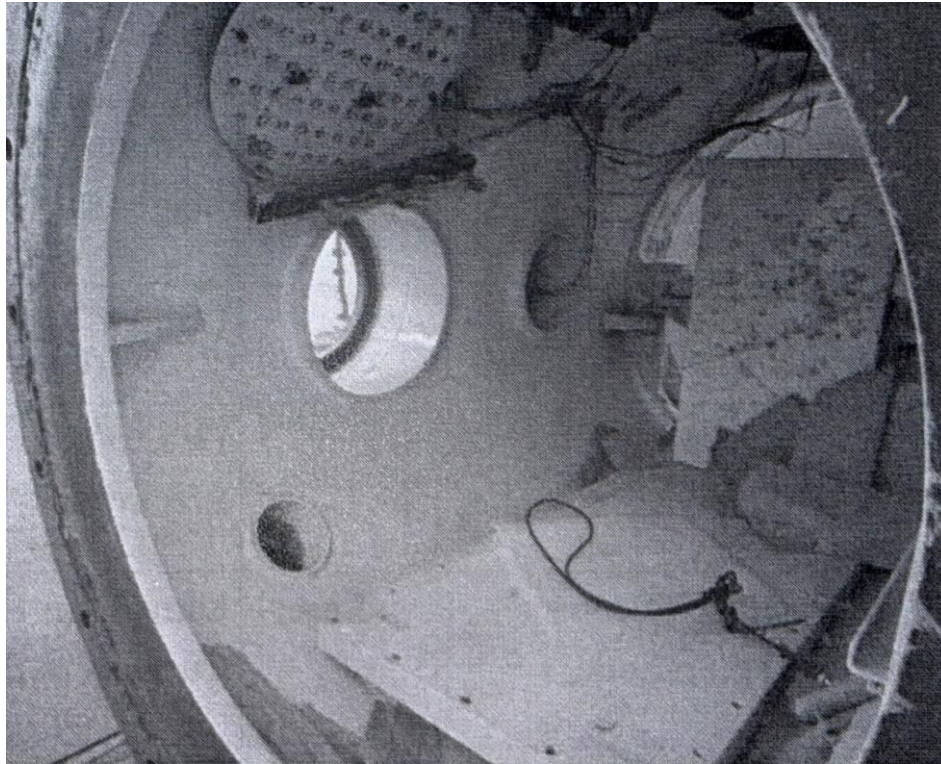


Figure 35 – Commercial velocity measurement setup.

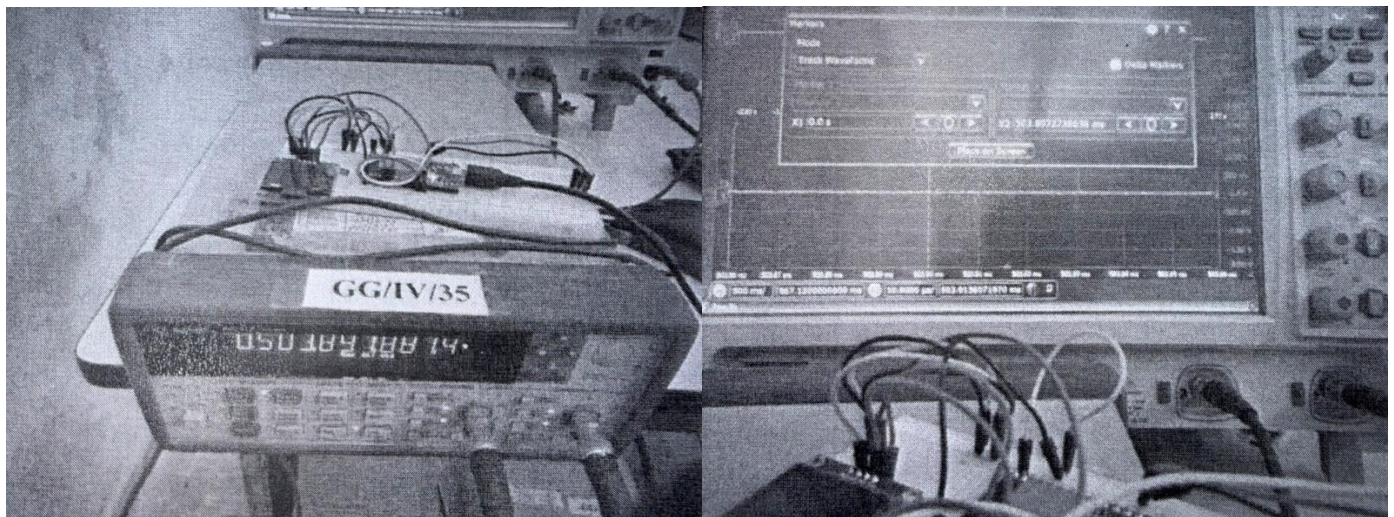


Figure 36 – Oscilloscope and Comparator reading

Qualitative Analysis of Architectural Differences

Table 17 highlights the structural, economic, and performance parameters that differentiate the two systems.

Table 17: Comparison between Developed PVMS and Commercial Projectile Velocity Measurement System

Parameter	Developed PVMS	Commercial System (Two-Stage Gas Gun Chronograph)
Measurement Principle	Optical Time-of-Flight	Optical Time-of-Flight
Controller	ESP32 Microcontroller	Dedicated High-Speed Electronics (FPGA/Precision Timing Circuitry)
Optical Sensors	IR Sensor + Laser-Photodiode	Ultra-Thin High-Speed Laser Curtains & Photodiodes
Timing Method	ESP32 Microsecond Timer (Microsecond Resolution)	Gigahertz Precision Timing (Nanosecond Resolution)
Display	Integrated Onboard OLED Display	Digital Chronograph / High-Bandwidth Oscilloscope
Operating Velocity Range	Low Speed (up to 10 m/s)	High Speed (up to 1500 m/s)
Intended Application	Educational Prototype & Laboratory Testing	Professional Ballistic & Aerospace Impact Testing
Cost	Extremely Low (Cost-Effective)	Exceptionally High (Industrial/Military Grade)

ne core divergence between the two systems lies in their hardware constraints rather than their underlying physics. The commercial device utilizes specialized, low-latency sensors and high- frequency clock counters to capture transient ballistic events. When a projectile travels at 505.88 m/s, a minor microsecond delay in sensor activation introduces catastmphic measurement errors. Conversely, the developed PVMS demonstrates that highly repeatable. stable velocity measurements can be achieved at a fraction of the Cost using off-the-shelf components for low- speed testing JOm/s. The microsecond-level timing of the ESP32 introduces negligible ermr at these lower speeds, as proven by the tight 0.07m/s variance shown in Table 16.

9.7 Section Conclusion and Future Outlook

In summary, while the commercial system and the developed PVMS operate at fundamentally different velocity scales, both successfully validate the reliability of the optical time-of-flight (TOF) measurement technique. The commercial chronograph provides the precision necessary for high-impact ballistic research, whereas the developed PVMS serves as an affordable and highly effective proof-of-concept for laboratory and educational environments.

The current PVMS framework is highly scalable. By upgrading the existing infrared and photodiode sensors to high-speed, sub-microsecond response photodiodes, incorporating dedicated analog comparator circuits to eliminate signal processing delays, and utilizing hardware-level timers for higher timing accuracy, the system can be adapted to measure significantly higher projectile velocities. This successful implementation establishes a robust hardware foundation for future enhancements, including the integration of machine learning algorithms for real-time projectile velocity estimation, trajectory analysis, and impact prediction.

Chapter 10 - Results and Discussion

10.1 Test Results Summary

The complete set of confirmed real test results across helmet sizes and impact angles, used both for application calibration and as the real-data anchor for ML model training, is reproduced below from Chapter 3.

Table 14: Summary of Confirmed Test Results

Sample	Angle (deg)	HIC	BrIC
Small-1	0	119.69	0.067
Small-2	0	114.07	0.242
Medium-1	0	132.21	0.325
Medium-2	0	152.24	0.763
Large-1	0	102.35	0.707
Large-2	0	159.78	0.298
Small-3	25	62.45	0.111
Small-4	45	53.44	0.171
Medium-3	25	46.17	0.112
Medium-4	45	26.1	0.098
Large-3	25	35.53	0.076
Large-4	45	9.42	0.054

10.2 Application vs DIAdem Accuracy

After applying correction factors of 1.4136 (HIC, ballistic), 1.867 (HIC, drop), and 0.5115 (BrIC, ballistic), the following representative comparison was obtained:

Table 15: Application Accuracy Compared to NI DIAdem

Test	HIC Python	HIC DIAdem	Error %	BrIC Python	BrIC DIAdem	Error %
1	119.5	119.69	0.2%	0.067	0.067	0.0%
2	132.0	132.21	0.2%	0.325	0.325	0.0%
3	159.6	159.78	0.1%	0.763	0.763	0.0%

10.3 Effect of Helmet Type on HIC

At 0-degree impact, HIC values do not show a strictly monotonic relationship with helmet size across the two samples tested per size (Small: 119.69, 114.07; Medium: 132.21, 152.24; Large: 102.35, 159.78), indicating substantial test-to-test variability at this small sample size. However, at the 25-degree and 45-degree angled impacts, the trend is clearer: HIC decreases consistently from Small to Medium to Large at each angle (25 degrees: 62.45 to 46.17 to 35.53; 45 degrees: 53.44 to 26.1 to 9.42), suggesting that larger helmet shells provide measurably better protection once the impact is not purely direct.

10.4 Effect of Impact Angle

Across all three helmet sizes, both HIC and BrIC decrease substantially as impact angle increases. For the Large helmet, HIC drops from 102.35-159.78 at 0 degrees to 35.53 at 25 degrees to 9.42 at 45 degrees - an order-of-magnitude reduction. This is consistent with the physical expectation that oblique impacts transfer a smaller fraction of the projectile's kinetic energy normal to the helmet surface, and that the helmet shell deflects rather than fully absorbs the impact at higher angles.

10.5 ML Model Performance

As discussed in Chapter 8, three machine learning algorithms were evaluated for predicting HIC and BrIC values. Among them, the Gradient Boosting model achieved the highest prediction accuracy, with R^2 values of 0.9384 for HIC and 0.8651 for BrIC. Based on these results, Gradient Boosting was selected as the primary prediction model integrated into the application.

10.6 Integration of the PVMS Companion Module

The successful validation of the Projectile Velocity Measurement System (PVMS) prototype in Chapter 8 directly supports the real-world deployment of the developed prediction application. By accurately capturing low-speed laboratory projectile data with minimal measurement variance, the PVMS provides a reliable and automated alternative to manual velocity input.

When compared to the industrial-grade two-stage light-gas gun chronograph used at TBRL, which operates at hypervelocities averaging **504.62 m/s**, the developed PVMS demonstrates that the optical time-of-flight (TOF) principle can be effectively implemented using affordable, off-the-shelf components for localized laboratory testing. Automating the velocity measurement process eliminates human input errors and creates a seamless data acquisition pipeline for the Gradient Boosting machine learning model, enabling highly accurate Head Injury Criterion (HIC) and Brain Injury Criterion (BrIC) predictions with an **R^2 score of 0.9384**, as discussed in Section 9.5.

Chapter 11 - Conclusion

11.1 Summary of Work

During the industrial training, I gained practical experience in the field of biomechanical impact analysis and helmet testing by working with the Hybrid III head-neck assembly and the associated sensor instrumentation. I studied the complete data acquisition and analysis pipeline used for ballistic helmet testing and developed a detailed understanding of Head Injury Criterion (HIC), Brain Injury Criterion (BrIC), signal processing techniques, and sensor behaviour.

As part of the training, I worked on a Python-based desktop application used for automated biomechanical analysis. My contribution involved understanding the software architecture, validating the HIC and BrIC calculations against NI DIAdem, analysing signal processing methods, studying machine learning models for injury prediction, and documenting the complete workflow.

I also designed and partially prototyped a compact Projectile Velocity Measurement System (PVMS) using an ESP32 microcontroller, optical sensing gates, and an OLED display. The prototype successfully demonstrated the optical time-of-flight principle for measuring the velocity of low-speed objects and was experimentally validated through repeated laboratory testing. The developed PVMS serves as a proof-of-concept for future integration with the prediction application to automate projectile velocity input.

11.2 Key Lessons Learned

The industrial training provided several practical insights that complemented the theoretical concepts studied during the academic curriculum:

- Small differences in filter implementation, or even the choice of whether to filter at all, can produce large and inconsistent differences in HIC and BrIC values because the nonlinear exponent of 2.5 in the HIC formula amplifies what appear to be negligible discrepancies. Calibrating a correction factor against a fixed, well-defined processing pipeline (raw, unfiltered data) proved more reliable than attempting to exactly replicate a black-box reference filter.
- The resultant of a multi-axis signal must be computed per-sample before finding its maximum, not from the per-axis peak values, since the components do not generally peak at the same instant.
- Angular acceleration derived by numerical differentiation of raw angular velocity data is dominated by noise; filtering before differentiation (for display purposes only) produces a physically meaningful graph without affecting any numeric injury-metric output.
- Data augmentation using physics-based synthetic records anchored to real measurements, including angle-dependent attenuation observed directly in the real dataset, is more effective than random augmentation for training small-dataset regressors.

11.3 Limitations

- Correction factors for HIC and BrIC are based on a limited number of confirmed test comparisons. More tests, particularly at additional angles such as 18 degrees and across the HSC ammunition type, are needed to confirm their stability.
- The prediction module includes a Patka option for demonstration purposes; however, corresponding experimental validation data are currently unavailable.
- The ML model was trained on real records augmented synthetically; performance on genuinely new test conditions outside the observed angle and helmet-size range may be lower than the cross-validation accuracy suggests.

11.4 Future Work

- Collect more test data across a wider range of conditions to retrain the ML model on real records without requiring synthetic augmentation.
- Implement Nij (Neck Injury Criterion) calculation using the Force and Moment channels already being recorded.
- Add real-time data acquisition mode to allow the application to record and analyse signals as a test is conducted, rather than from saved CSV files.
- Investigate whether implementing the exact ISO 6487 Channel Frequency Class filter coefficients (rather than relying on a correction factor) further improves agreement with NI DIAdem.
- Integrate the PVMS hardware directly with the desktop application through serial communication to enable automatic real-time velocity input for machine learning predictions.

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